LECTURE · WEDNESDAY, OCTOBER 28, 2026

Clustering validation

Choosing clusters when there are no labels to check against

Kai Wycik · BCB726: Machine Learning for Computational Biology. Fall 2026

ABOUT ME

Hi, I’m Kai (or just "K")

B.Sc. Information Systems
University of Cologne

→

M.A. Statistics
Columbia University

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Ph.D. student, STOR
UNC Chapel Hill

OUTSIDE OF RESEARCH

- Philosophy and critical theory
- Video games, and competitive Pokémon VGC
- Guitar

Graduating from Columbia

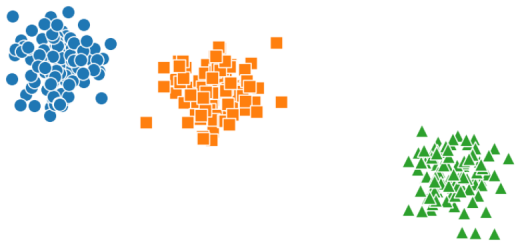
Pokémon VGC at EUIC 2024

Playing guitar in a concert

MOTIVATION

Clustering finds groups without labels

Supervised: labels are given



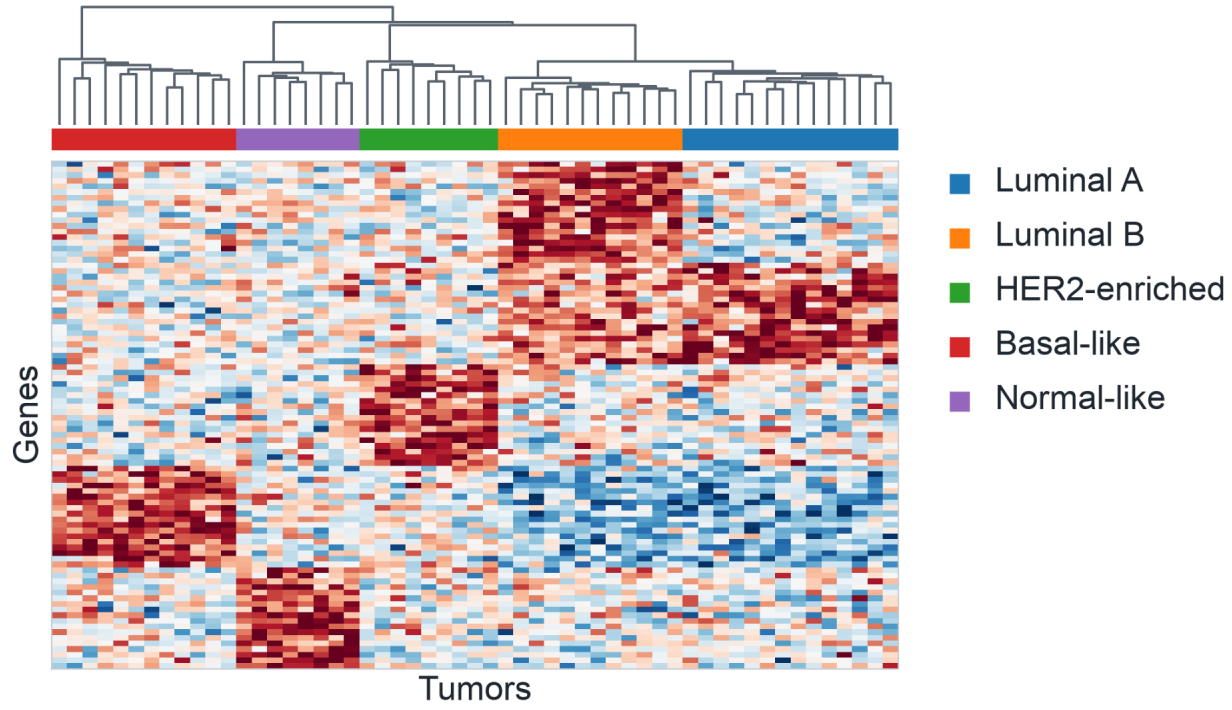
Each cell comes with a label, such as a known cell type. We learn a rule that predicts it and test the rule on labeled cells we held back.

Unsupervised: no labels

The same cells without labels. We ask whether they fall into groups, and how many groups there are.

EXAMPLE: CANCER

Breast cancer subtypes came from clustering



Schematic with **simulated** values: genes in rows, tumors in columns, red for high and blue for low expression.

Perou et al. (2000) grouped breast tumors by hierarchical clustering of their gene expression. Sørlie et al. (2001) refined the groups and showed that patients in different groups had different survival.

The subtypes (luminal A, luminal B, HER2-enriched, basal-like, normal-like) are now part of how breast cancer is classified. The PAM50 test assigns a new tumor to one of them from 50 genes (Parker et al., 2009).

Perou et al., Nature 2000 · Sørlie et al., PNAS 2001 · Parker et al., J Clin Oncol 2009

SOME MOTIVATIONAL FOOD FOR THOUGHT

Marker genes, differential expression and cell-type proportions are all computed on groups we find via clustering.

They are only as reliable as the clusters themselves. How do we check a clustering when there is nothing to compare it with?

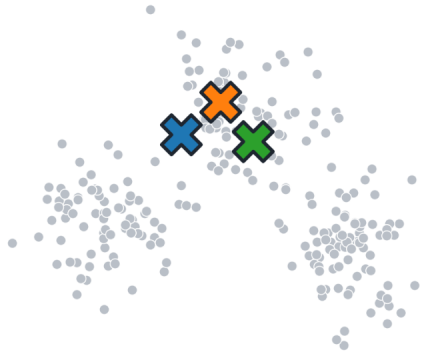
01

How clustering algorithms work

k-means, hierarchical clustering and spectral clustering

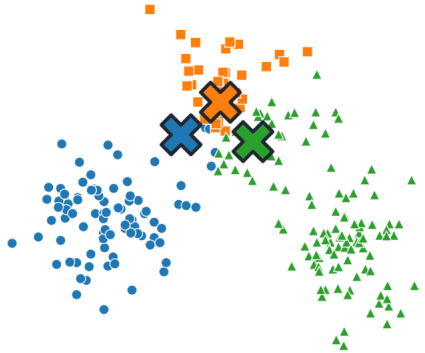
K-MEANS

k-means repeats two steps until it settles



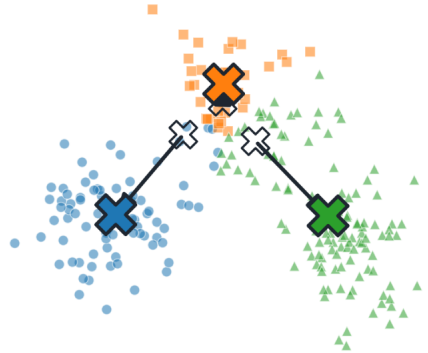
Step 1

Pick k starting centers. Here k = 3, all in one group.



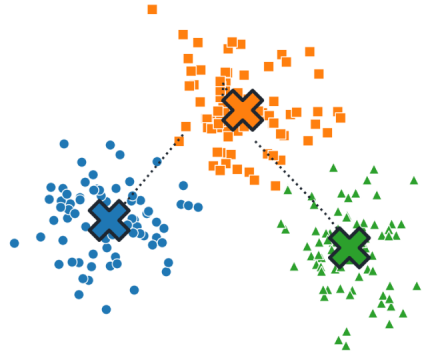
Step 2

Assign every point to its nearest center.



Step 3

Move each center to the mean of its points.



Step 4

Repeat 2 and 3 until no point changes cluster. Here it took 4 rounds.

The goal: k centers with the smallest total squared distance from each point to its center.

K-MEANS

What k-means assumes

Users choose k

The algorithm returns exactly k clusters, whether or not the data have k groups.

Groups are round

Every point joins its nearest center, so the boundaries between clusters are straight lines. Stretched, curved or nested groups get cut apart.

The start matters

Different starting centers can end in different answers. Software runs several starts and keeps the best one (n_init in scikit-learn).

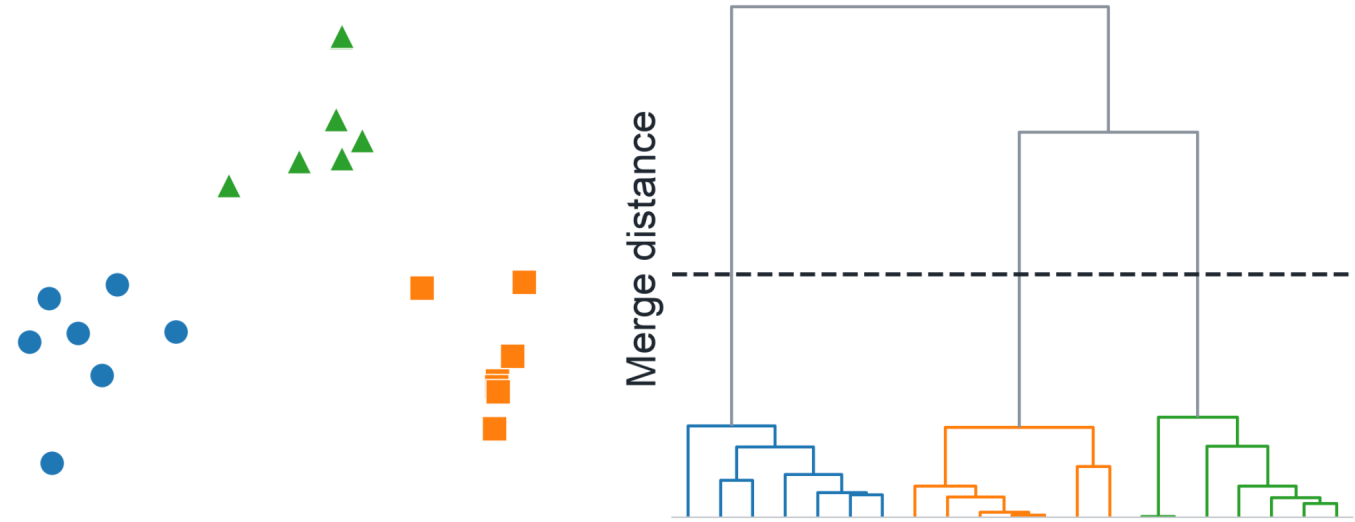
The next two algorithms drop the round-groups assumption. Neither removes the need to choose k.

HIERARCHICAL CLUSTERING

Hierarchical clustering builds a tree

1. Start with every point as its own cluster.
2. Merge the two closest clusters.
3. Repeat until one cluster is left.
4. Cut the tree at a height to get k clusters.

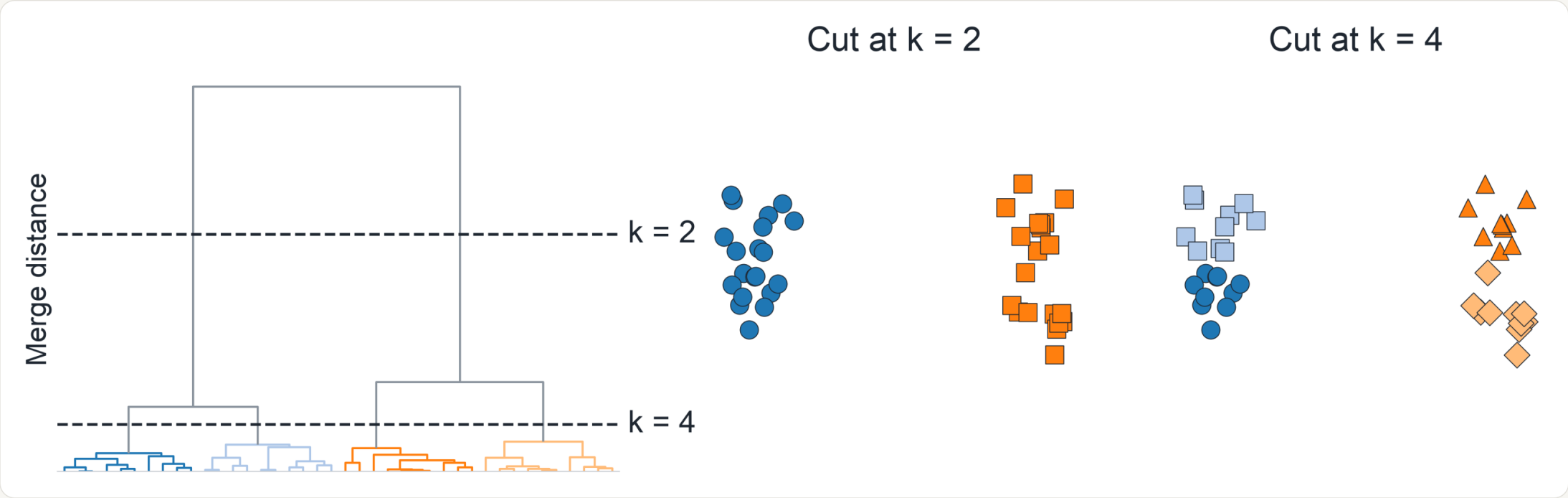
The linkage decides what “closest” means: single (nearest pair of points), complete (farthest pair), average, or Ward (smallest increase in within-cluster variance).



Ward linkage. Each merge is drawn at the distance where it happened; cutting at the dashed line gives three clusters.

HIERARCHICAL CLUSTERING

One tree holds every k

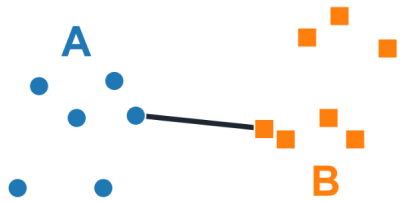


Cutting high gives two clusters. Cutting lower splits each of them again, so the four clusters sit inside the two. Cell types are often nested the same way: T cells contain CD4 and CD8 T cells.

HIERARCHICAL CLUSTERING

Linkage defines the distance between clusters

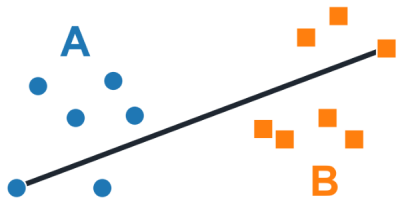
Between two points we use a distance, usually Euclidean: $d(a, b) = \|a - b\|$



Single linkage

$$d(A, B) = \min_{a \in A,\ b \in B} \|a - b\|$$

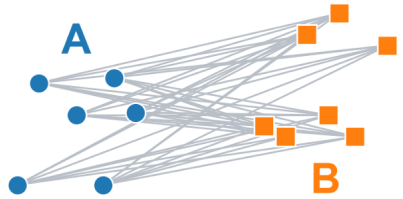
The closest pair of points. Follows chains of points.



Complete linkage

$$d(A, B) = \max_{a \in A,\ b \in B} \|a - b\|$$

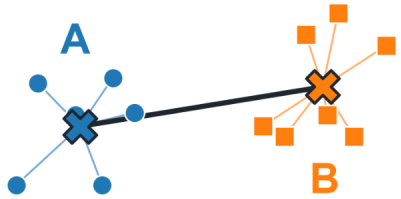
The farthest pair of points. Keeps clusters compact.



Average linkage

$$d(A, B) = \frac{1}{|A|\,|B|} \sum_{a \in A} \sum_{b \in B} \|a - b\|$$

The mean over all pairs of points.



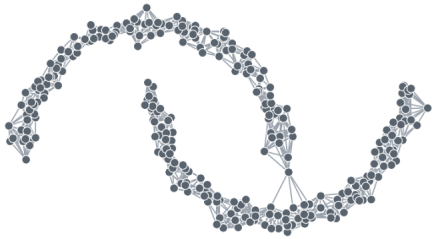
Ward linkage

$$\Delta(A, B) = \frac{|A|\,|B|}{|A| + |B|} \|\bar{a} - \bar{b}\|^2$$

Increase in within-cluster sum of squares. Bars mark the means.

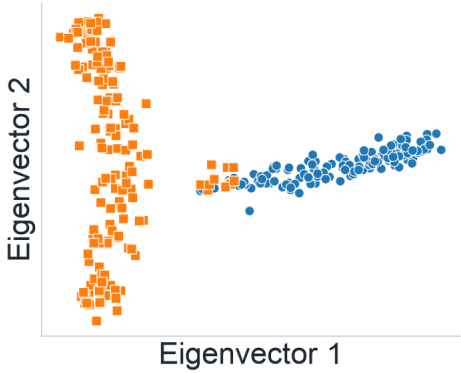
SPECTRAL CLUSTERING

Spectral clustering uses a neighbor graph



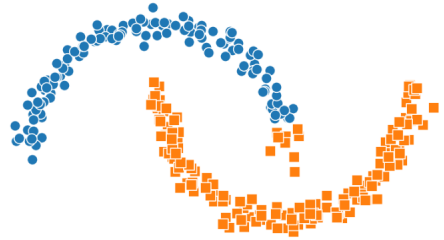
Step 1

Connect each point to its 10 nearest neighbors.



Step 2

Take the eigenvectors of L with the smallest eigenvalues.



Step 3

Run k-means on the new coordinates.

$$L = D - W$$

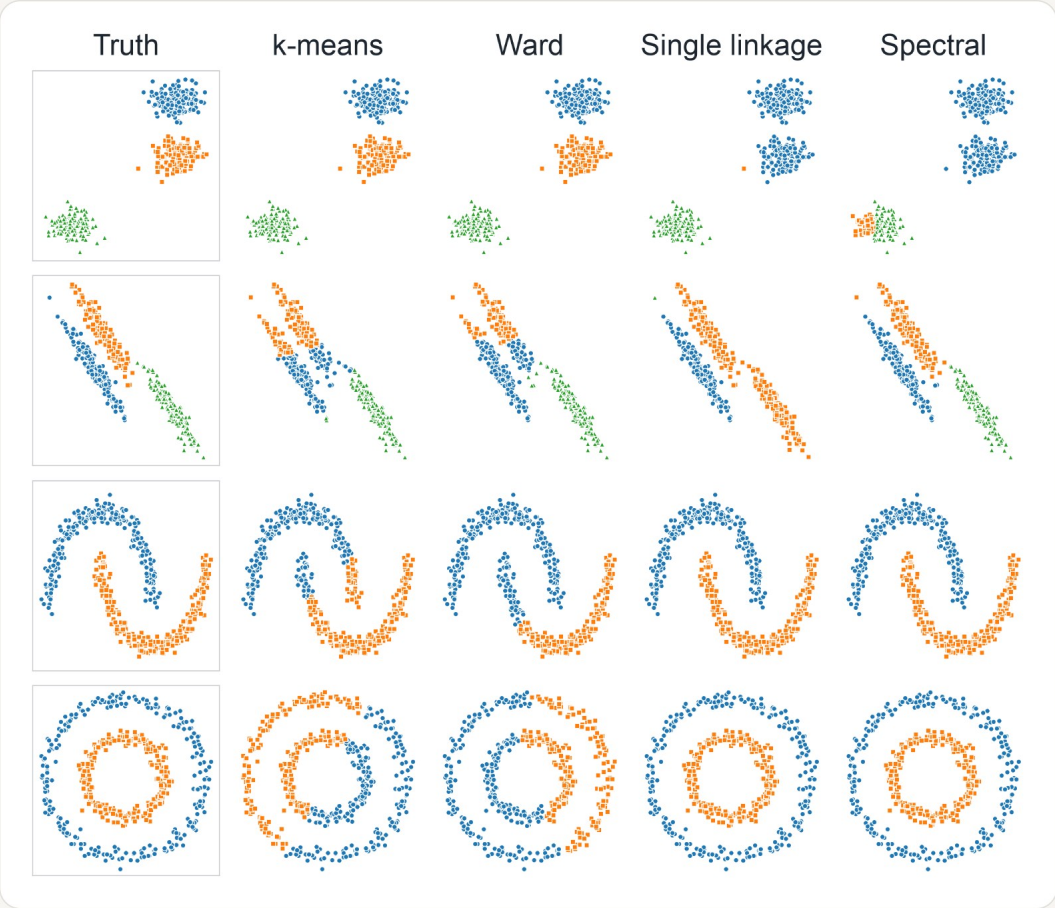
W: 1 between neighbors, 0 otherwise. D: number of neighbors, on the diagonal.

$$f^{\mathrm{T}} L f = \frac{1}{2} \sum_{i,j} w_{ij} \left(f_i - f_j\right)^2$$

Small when f changes little along edges. The eigenvectors with the smallest eigenvalues do this best, so neighbors get similar coordinates.

COMPARING METHODS

No algorithm works on every shape



Four simulated data sets, each method given the right k. The first column shows the groups the data were simulated from.

- k-means and Ward find the round groups and cut the stretched, curved and nested ones.
- Single linkage follows chains of points. It finds the moons and rings, but one stray point makes it merge two round groups.
- Spectral clustering finds the stretched groups, moons and rings, but merges two round groups and splits the third.

The right algorithm depends on the shape of the groups, and in real data we do not know that shape.

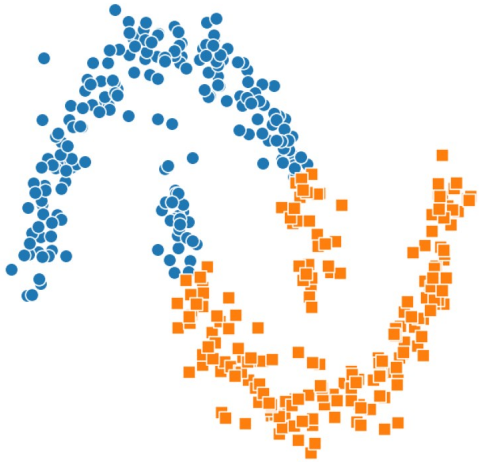
02

Choosing settings without labels

Hyperparameters, the number of clusters, and validity indices

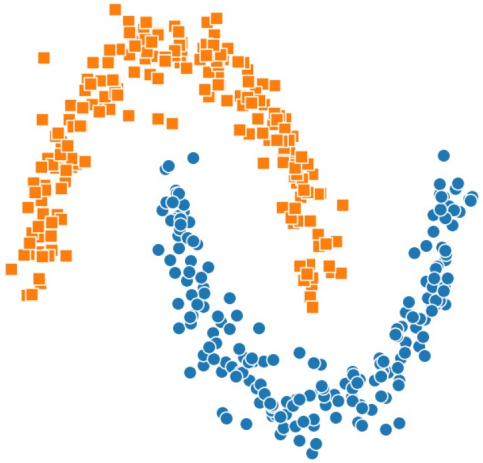
HYPERPARAMETERS

One setting, three different answers



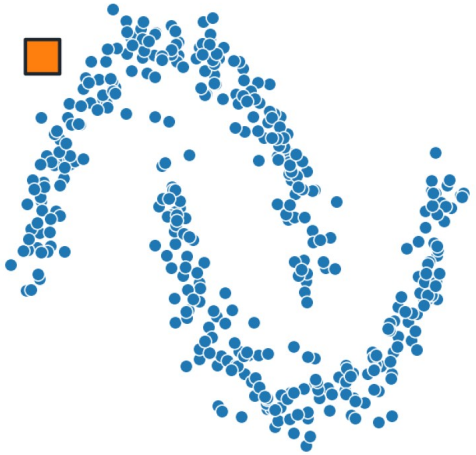
γ = 0.3

A straight cut through both moons.



γ = 30

The two moons.



γ = 300

One stray point against the other 399.

$$w_{ij} = \exp(-\gamma \, \|x_i - x_j\|^2)$$

Spectral clustering with k = 2 every time. Every pair of points gets this Gaussian weight in W, and γ sets how fast it drops with distance.

PREPROCESSING

How a published atlas chose its clusters

> “Finally, we generated a lower dimensional representation of the data by including the first 50 dimensions of the singular value decomposition of this TF-IDF-transformed matrix. This representation was then used as input for the Rtsne package in R. To identify clusters of cells in this two dimensional representation of the data, we used the Louvain clustering algorithm implemented in Seurat. Resolution and K parameters for Louvain clustering were chosen for each major cluster to produce reasonable groupings of cells that are well-separated in each t-SNE embedding.”

Cusanovich et al., Cell 2018, STAR Methods (citations omitted). 81,173 cells from 13 mouse tissues.

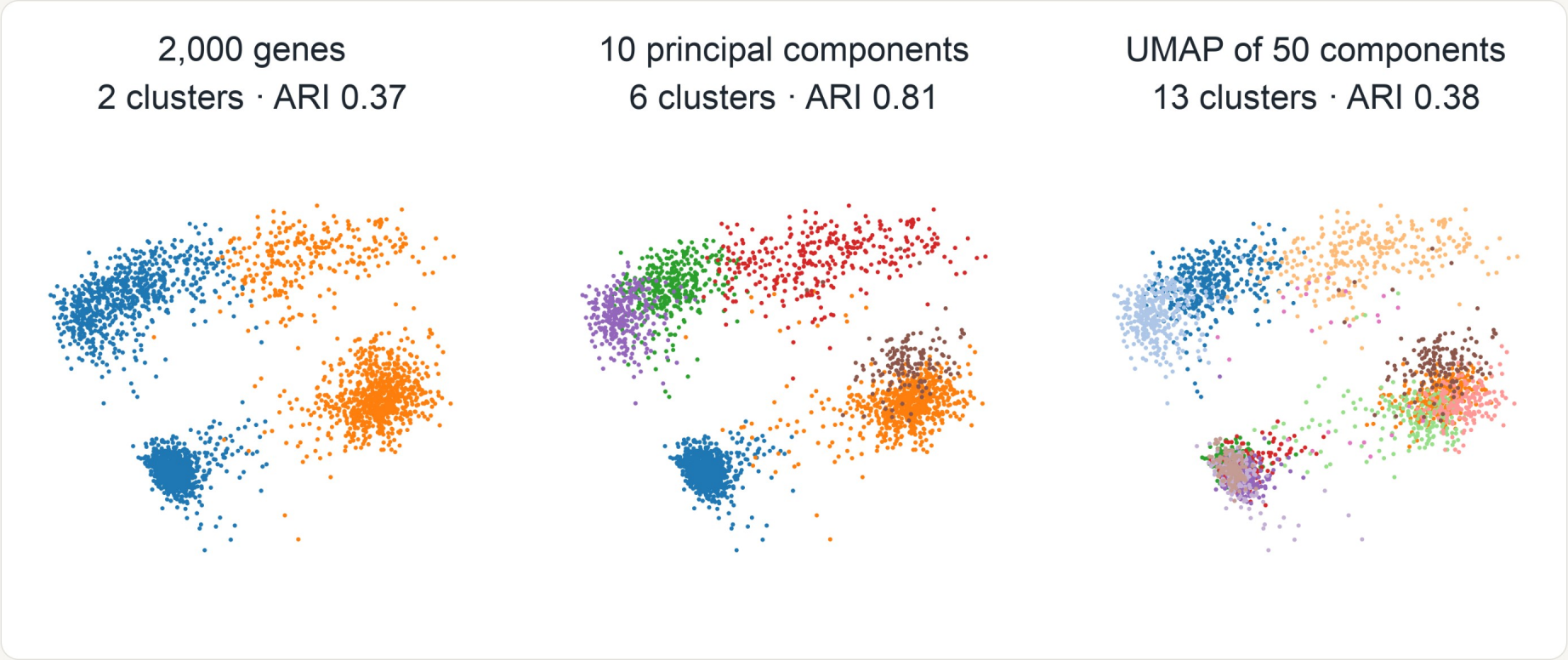
The paragraph fixes three choices: 50 components, a two-dimensional t-SNE, and Louvain with its resolution and K.

The settings were picked so the clusters look well separated in the t-SNE. The paper reports neither the perplexity nor the resolution.

This is common. Methods sections list the pipeline but seldom say how it was chosen.

PREPROCESSING

Same algorithm, same resolution, different input

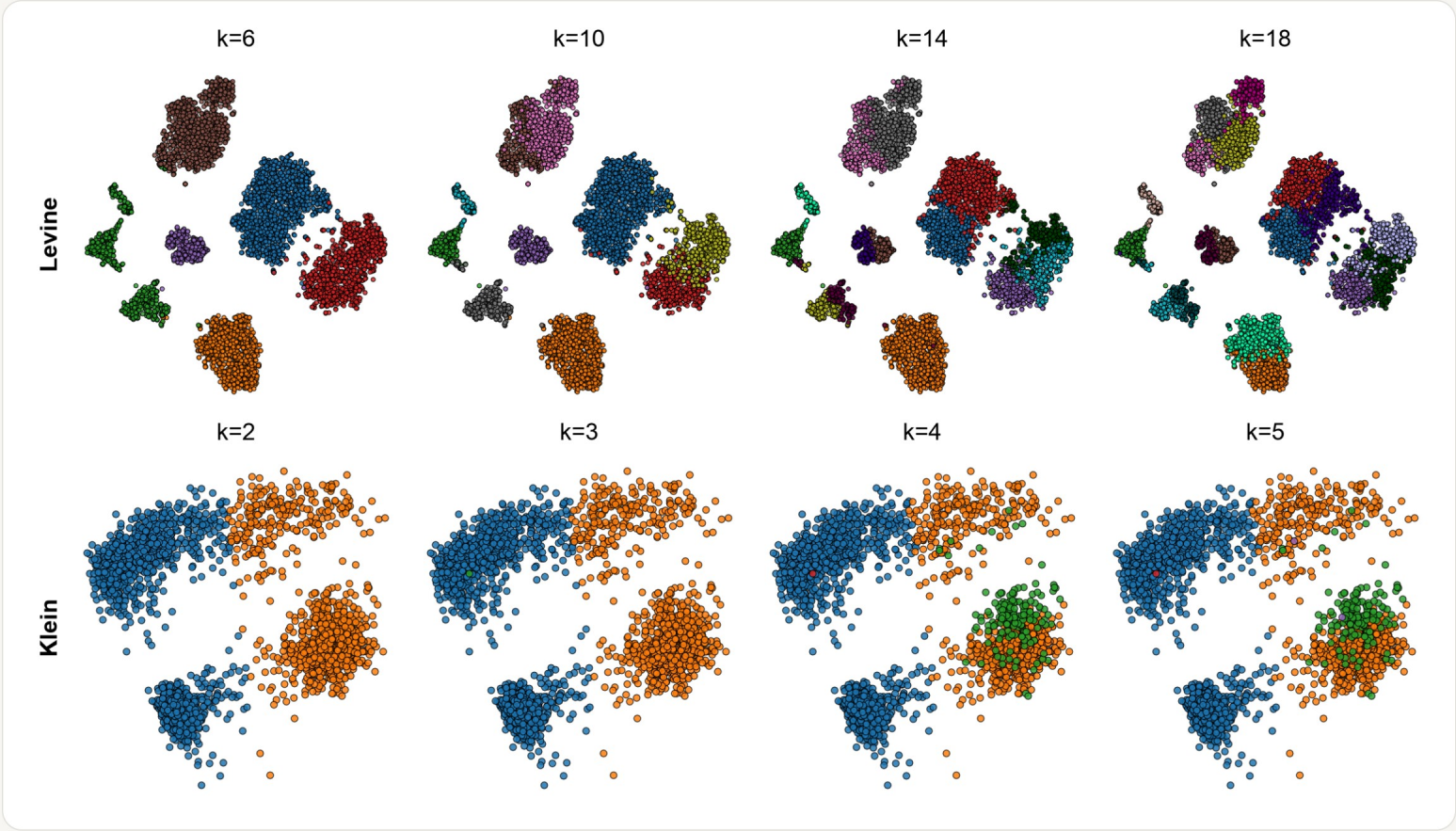


Leiden, resolution 0.5, on the Klein cells. Same PCA layout in every panel. ARI against collection day.

Klein et al., Cell 2015

NUMBER OF CLUSTERS

The number of clusters changes the biology



k-means on two real data sets. Only k changes.

Top: Levine mass cytometry, k = 6, 10, 14, 18.

Bottom: Klein scRNA-seq, k = 2, 3, 4, 5.

Each panel would be reported as a different number of cell types or cell states.

Levine et al., Cell 2015 · Klein et al., Cell 2015 · figure from the CARVE manuscript

THE PROBLEM

Three central choices

Preprocessing

Normalization, the number of principal components, an embedding such as UMAP or t-SNE.

Algorithm

k-means, Ward, spectral, Leiden, and their settings: linkage, γ, number of neighbors.

Number of clusters

k, or the resolution for Leiden and Louvain.

For example, 3 normalizations × 4 inputs × 4 algorithms × 9 values of k give 432 candidate pipelines. A paper reports one of them.

Can one framework choose all three?

THE CORE DIFFICULTY

Clustering has no straight-forward answer

Supervised learning

Hold out part of the labeled data. Choose the settings whose predictions best match the held-out labels: cross-validation.

- Very well researched
- Well established practice.

Clustering

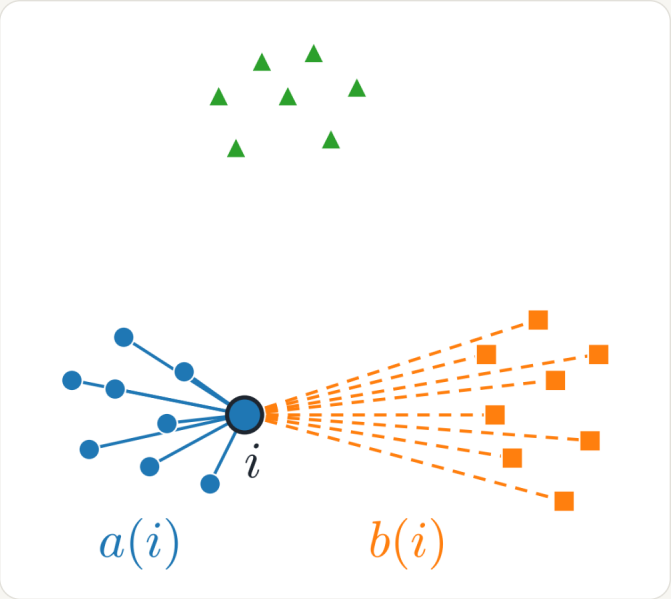
There are no labels to hold out. Any score has to come from the data and the clustering alone.

So how do we choose which algorithm and hyper-parameters to use?

No single rule for choosing these works for every method and every data set.

THE USUAL ANSWER: VALIDITY INDICES

Silhouette compares two distances per point



$$a(i) = \frac{1}{|C_i| - 1} \sum_{j \in C_i,\ j \neq i} d(i, j)$$

Mean distance from i to the other points in its own cluster.

$$b(i) = \min_{C \neq C_i} \frac{1}{|C|} \sum_{j \in C} d(i, j)$$

Mean distance from i to the points of the nearest other cluster.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i),\ b(i)\}}$$

Between –1 and 1. Near 1 when a(i) is much smaller than b(i).

$$S = \frac{1}{n} \sum_{i=1}^{n} s(i)$$

The silhouette score of the clustering. Choose the k with the highest S.

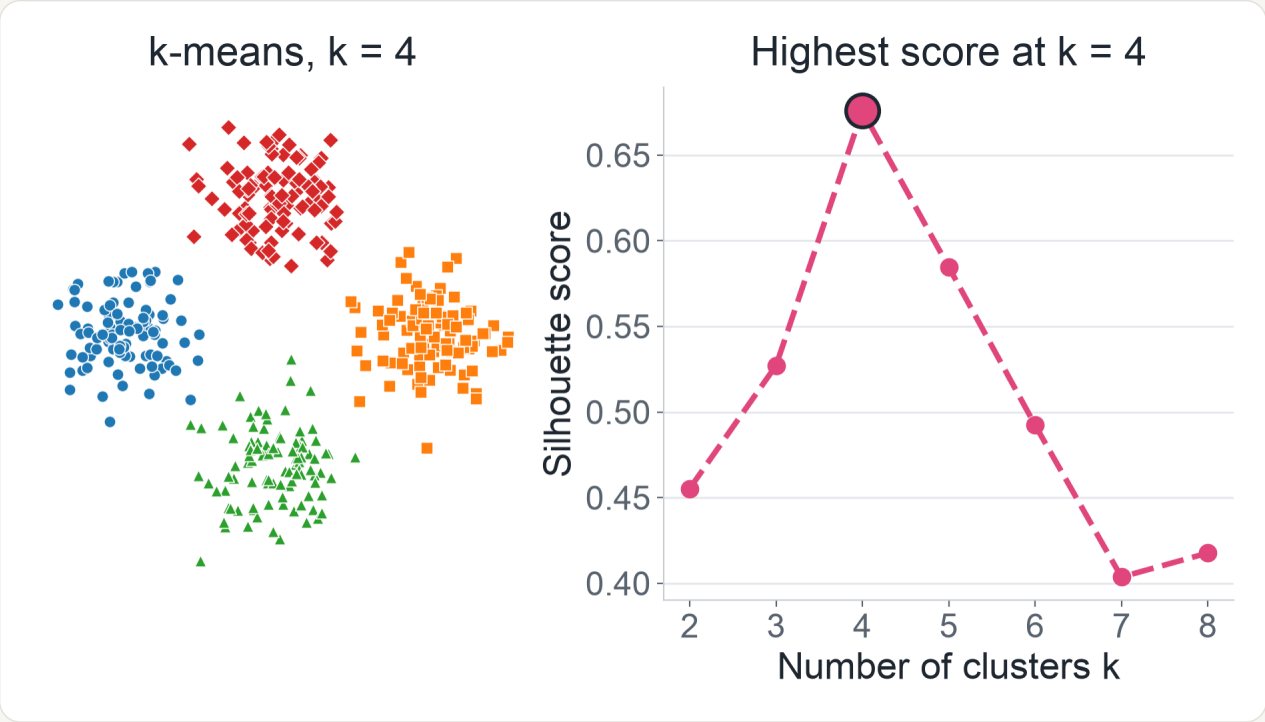
Other common indices

- Davies–Bouldin: spread within clusters relative to the distance between their centers. Lower is better.
- Calinski–Harabasz: spread between clusters relative to the spread within them. Higher is better.
- Gap statistic: how much tighter the clusters are than in random data with no groups.

Rousseeuw 1987 · Davies & Bouldin 1979 · Caliński & Harabasz 1974 · Tibshirani, Walther & Hastie 2001

EXAMPLE: SIMULATED DATA

Silhouette finds k on well-separated groups



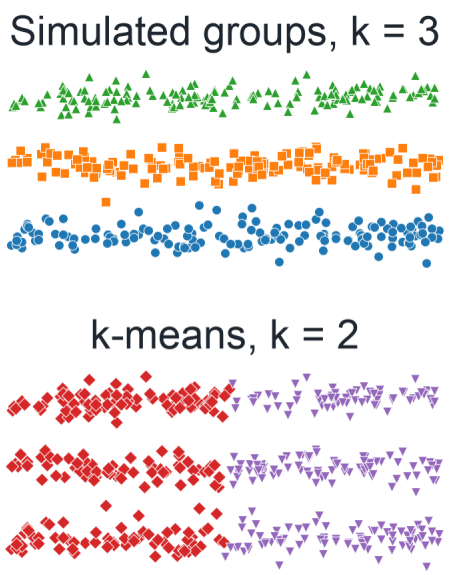
Four simulated groups. k-means for k = 2 to 8, and the silhouette score of each result.

The score is highest at k = 4, the number of groups we simulated.

Here the groups are what silhouette rewards: compact and well separated.

EXAMPLE: SIMULATED DATA

Silhouette fails on long, parallel groups



Three long, parallel groups. k-means for k = 2 to 8, as on the previous slide.

The simulated groups score 0.12. The highest score, 0.51, goes to k = 2, which cuts every group in half.

All four indices reward compact, well-separated groups. Cells along a differentiation path, or on a continuum of states, are neither.

03

Stability and generalizability

Checking a clustering by resampling the data

VALIDATION

What it means to validate a clustering

Validation checks whether a clustering describes structure in the cells. A clustering that changes with the sample, the algorithm or its settings is problematic in terms of scientific validity.

External

Compare with labels we trust, such as known cell types or collection days.

Rarely available for new data.

Internal

Score the clustering from the data alone, like the validity indices in the last section.

Necessitates assumptions about the underlying data.

Replication

Perturb the data and check whether the same clusters come back.

Stability and generalizability, the topic of this section.

VALIDATION

Results should hold up on new data

In the lab, we report something as a finding if and only if it replicates. Interpretable machine learning applies the same notion to what a model finds (Allen et al., 2023). For clustering this gives two questions.

Stability

If we cluster a different random subsample of the cells, do we get the same clusters?

Ben-Hur et al. 2002 · Monti et al. 2003

Generalizability

If we train a classifier on the clusters, can it predict the clusters of cells it has not seen?

Dudoit & Fridlyand 2002 · Tibshirani & Walther 2005

Neither question assumes a cluster shape, so both work with any clustering algorithm.

COMPARING TWO CLUSTERINGS

The adjusted Rand index measures agreement

The ARI compares two clusterings of the same points. It checks every pair of points: are the two together in both clusterings, or apart in both?

ARI = 1.00

Same groups, different names. The names do not matter.

ARI = 0.54

Two of the twelve points swapped groups.

ARI = −0.07

Unrelated groupings score about 0.

Top row: clustering A. Bottom row: clustering B. An ARI of 1 means identical groupings; values near 0 mean no more agreement than chance.

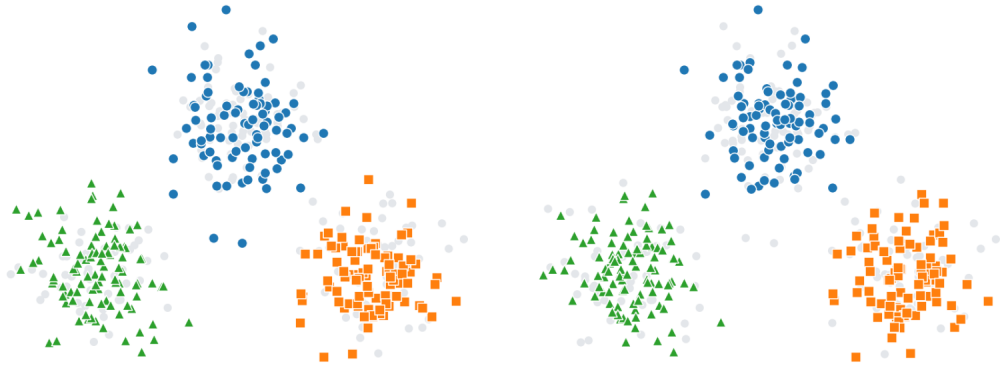
Hubert & Arabie, Journal of Classification 1985

STABILITY

Stability: do two subsamples agree?

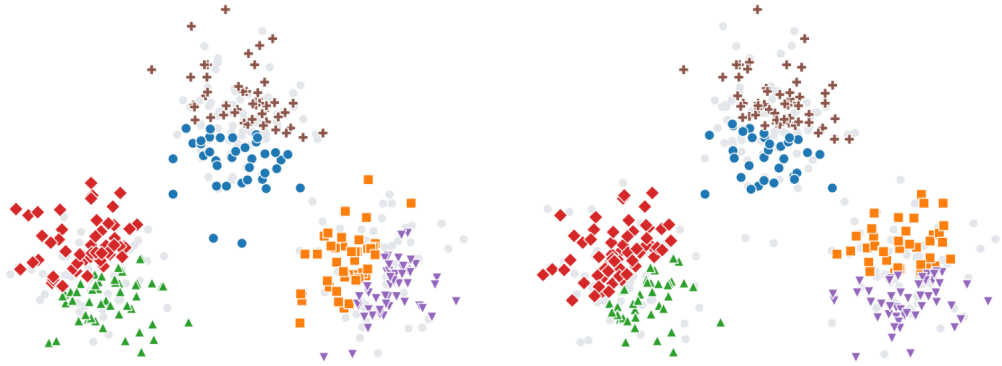
Cluster two random subsamples (61.8% of the points each) and compare them on the points they share.

k = 3 · ARI 1.00



Subsample 1 Subsample 2

k = 6 · ARI 0.77



Subsample 1 Subsample 2

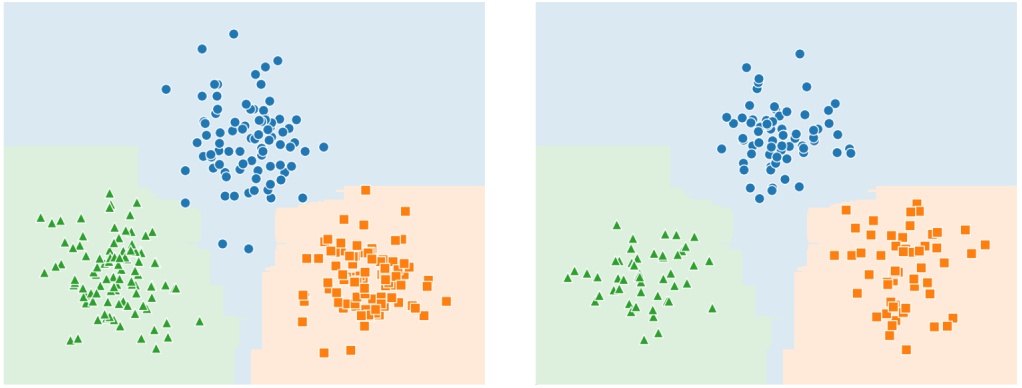
Light gray points were not drawn. With k = 6 the extra splits land in different places each time.

GENERALIZABILITY

Generalizability: predicting new cells

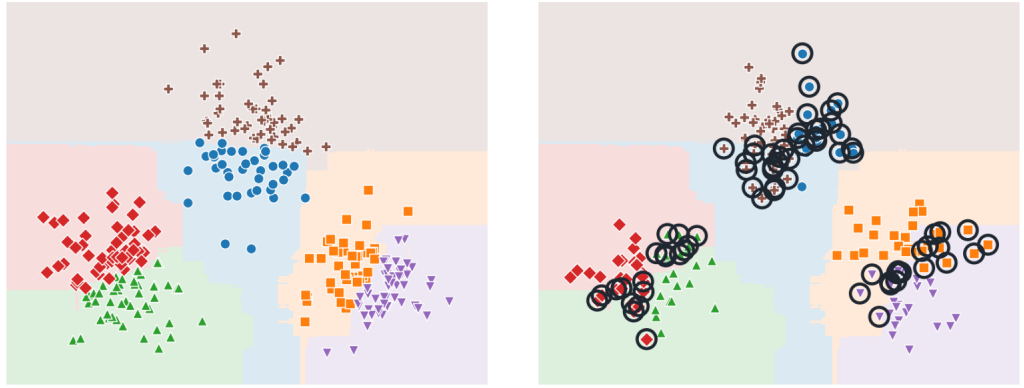
Cluster a subsample, train a random forest on its labels, predict the held-out points, and compare the predictions with a clustering of the held-out points.

k = 3 · ARI 1.00



Training subsample | Held-out points

k = 6 · ARI 0.44



Training subsample | Held-out points

Shading: the cluster the classifier predicts in each region. Circled: held-out points where the classifier and the clustering disagree (42% of them at k = 6).

A CAVEAT

Splitting cells into two halves is almost always stable and easy to predict.

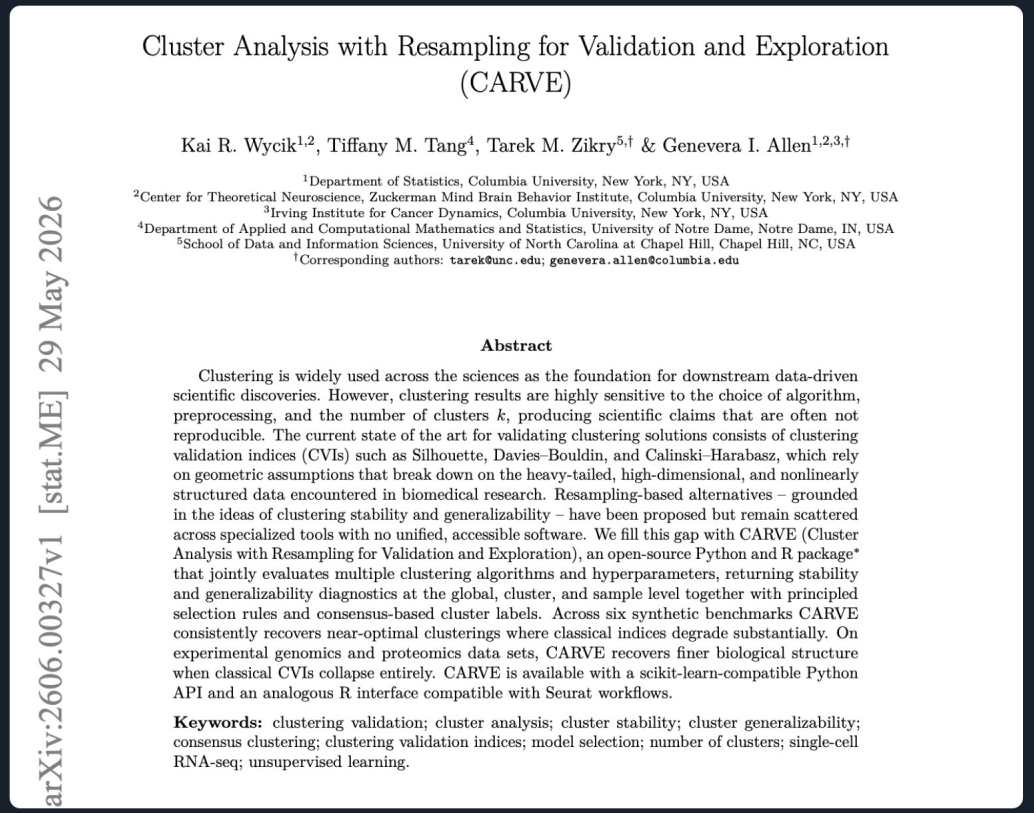
High scores are necessary for a clustering we can trust. They do not show that the clusters mean something biologically, so compare the candidates and use what you know about the cells.

04

CARVE

Cluster Analysis with Resampling for Validation and Exploration

Wycik, Tang, Zikry & Allen · arXiv:2606.00327



Cluster Analysis with Resampling for Validation and Exploration (CARVE)

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arXiv:2606.00327v1 [stat.ME] 29 May 2026

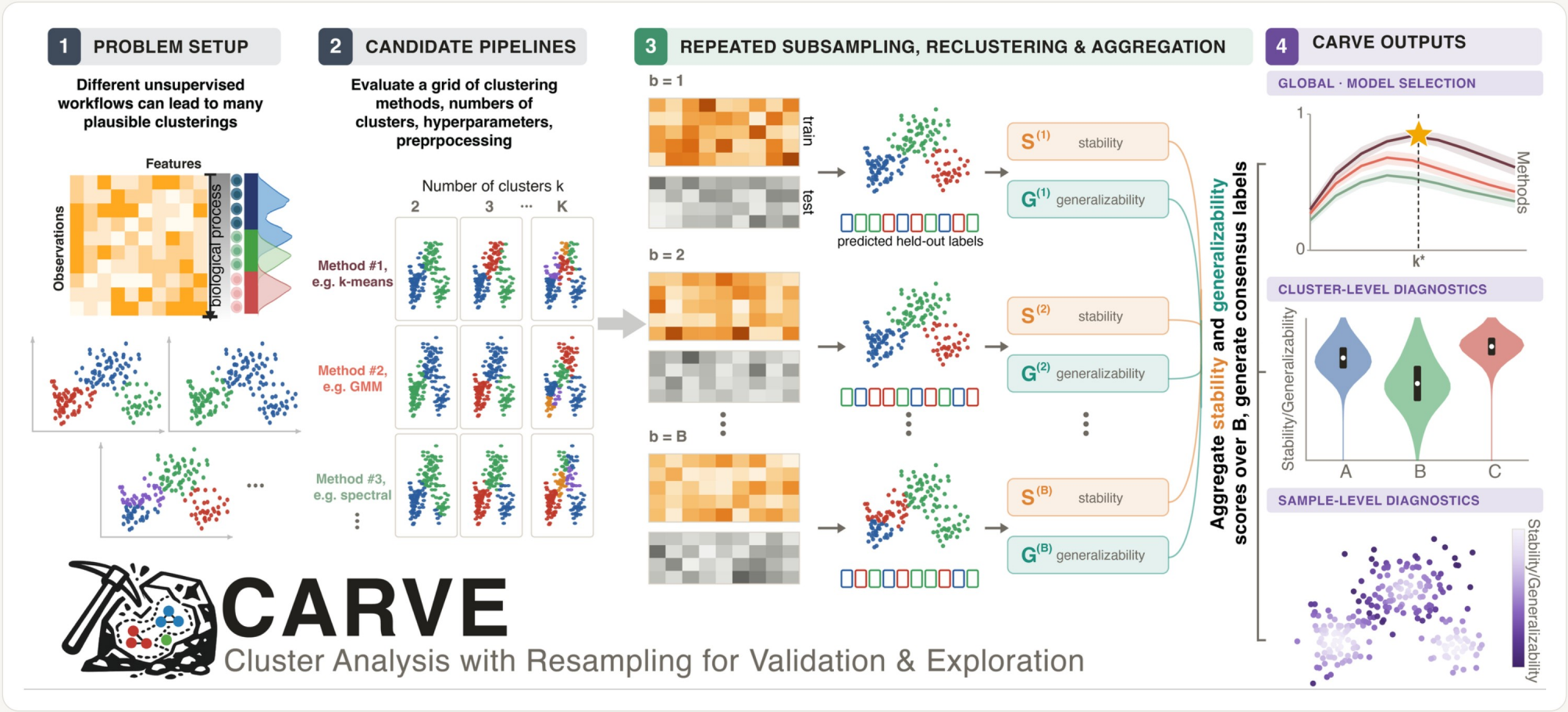
Abstract

Clustering is widely used across the sciences as the foundation for downstream data-driven scientific discoveries. However, clustering results are highly sensitive to the choice of algorithm, preprocessing, and the number of clusters k, producing scientific claims that are often not reproducible. The current state of the art for validating clustering solutions consists of clustering validation indices (CVIs) such as Silhouette, Davies–Bouldin, and Calinski–Harabasz, which rely on geometric assumptions that break down on the heavy-tailed, high-dimensional, and nonlinearly structured data encountered in biomedical research. Resampling-based alternatives – grounded in the ideas of clustering stability and generalizability – have been proposed but remain scattered across specialized tools with no unified, accessible software. We fill this gap with CARVE (Cluster Analysis with Resampling for Validation and Exploration), an open-source Python and R package* that jointly evaluates multiple clustering algorithms and hyperparameters, returning stability and generalizability diagnostics at the global, cluster, and sample level together with principled selection rules and consensus-based cluster labels. Across six synthetic benchmarks CARVE consistently recovers near-optimal clusterings where classical indices degrade substantially. On experimental genomics and proteomics data sets, CARVE recovers finer biological structure when classical CVIs collapse entirely. CARVE is available with a scikit-learn-compatible Python API and an analogous R interface compatible with Seurat workflows.

Keywords: clustering validation; cluster analysis; cluster stability; cluster generalizability; consensus clustering; clustering validation indices; model selection; number of clusters; single-cell RNA-seq; unsupervised learning.

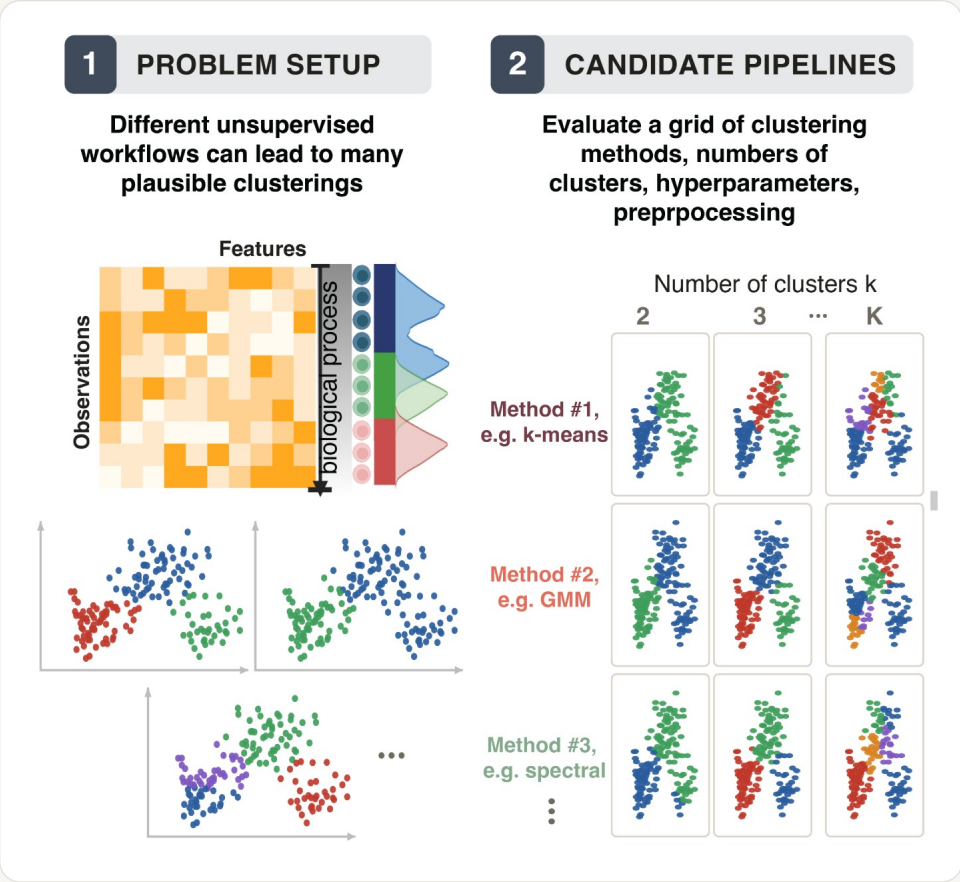
CARVE

CARVE scores every candidate clustering



Python and R packages: github.com/DataSlingers/CARVE

Input: data and candidate pipelines



A data matrix: one row per cell, one column per feature (genes, peaks, protein markers or principal components).

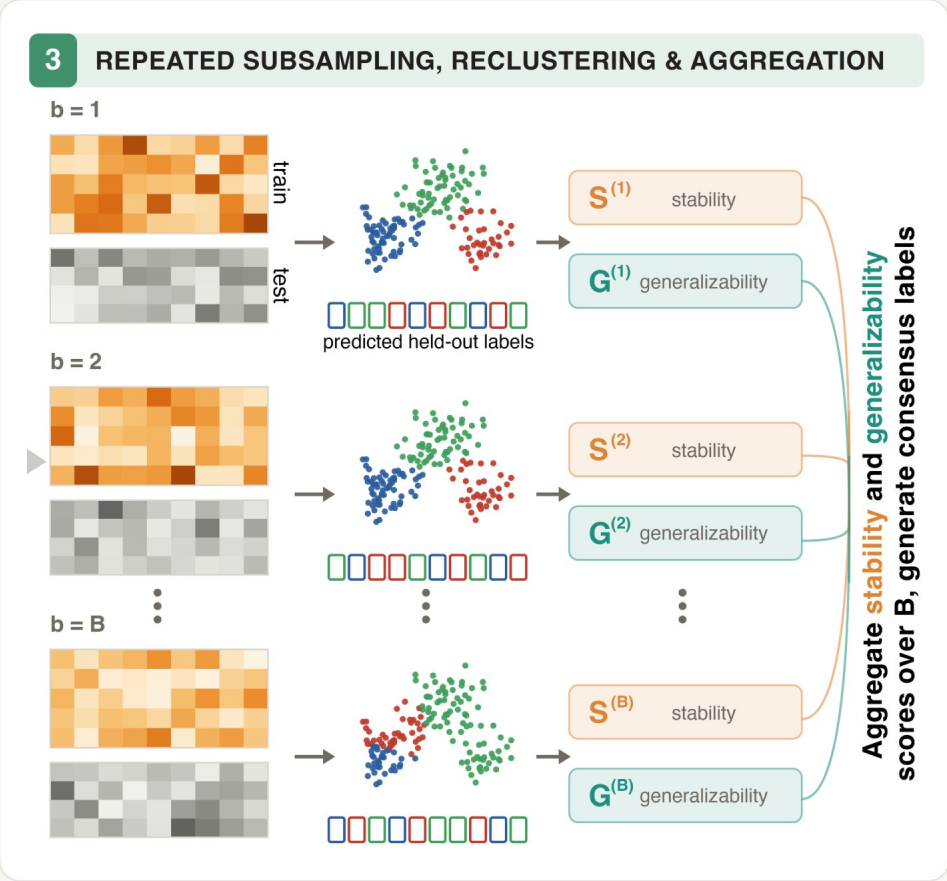
A grid of candidate pipelines. Each candidate is an algorithm with one value of k or resolution, its other settings, and optionally a preprocessing step.

So the grid can cover all three choices from earlier: preprocessing, algorithm and k.

In the case study: Ward and spectral clustering, each with k = 2 to 10, so 18 candidates.

CARVE · RESAMPLING

Repeat B times: subsample, recluster, score



1. Draw two subsamples, each with 61.8% of the cells. The cells outside the first subsample are the test set.
2. Cluster both subsamples and the test set with the candidate pipeline.
3. Score stability as the ARI between the two subsample clusterings on their shared cells.
4. Score generalizability: train a random forest on the first subsample's clusters, predict the test set, and compute the ARI with the test set's own clustering.

Defaults: B = 100 resamples, subsample fraction 0.618. Every candidate is scored on the same subsamples.

CARVE · OUTPUT

Scores at three levels

Global

Mean stability and generalizability of each candidate, with standard errors. Used to choose k and the algorithm.

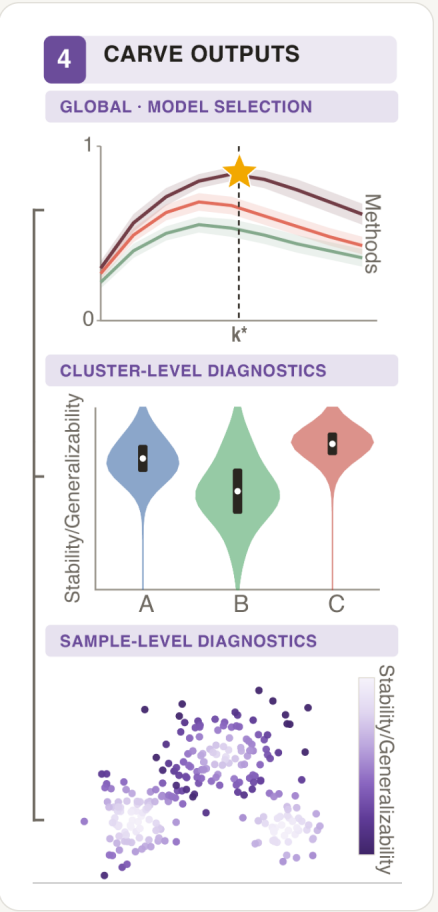
Cluster

Scores averaged over the cells of each cluster in the chosen solution. Shows which clusters hold up.

Cell

How consistently each cell lands in the same cluster, and how often its cluster is predicted correctly when it is held out.

Cell stability comes from the consensus matrix: for each pair of cells, the fraction of resamples that put them in the same cluster, out of those that drew both (Monti et al., 2003). Each cell's row is summarized by a Gini index, which is near 1 when the cell always lands with the same partners.



CARVE · SELECTION

Choosing k: the one-standard-error rule

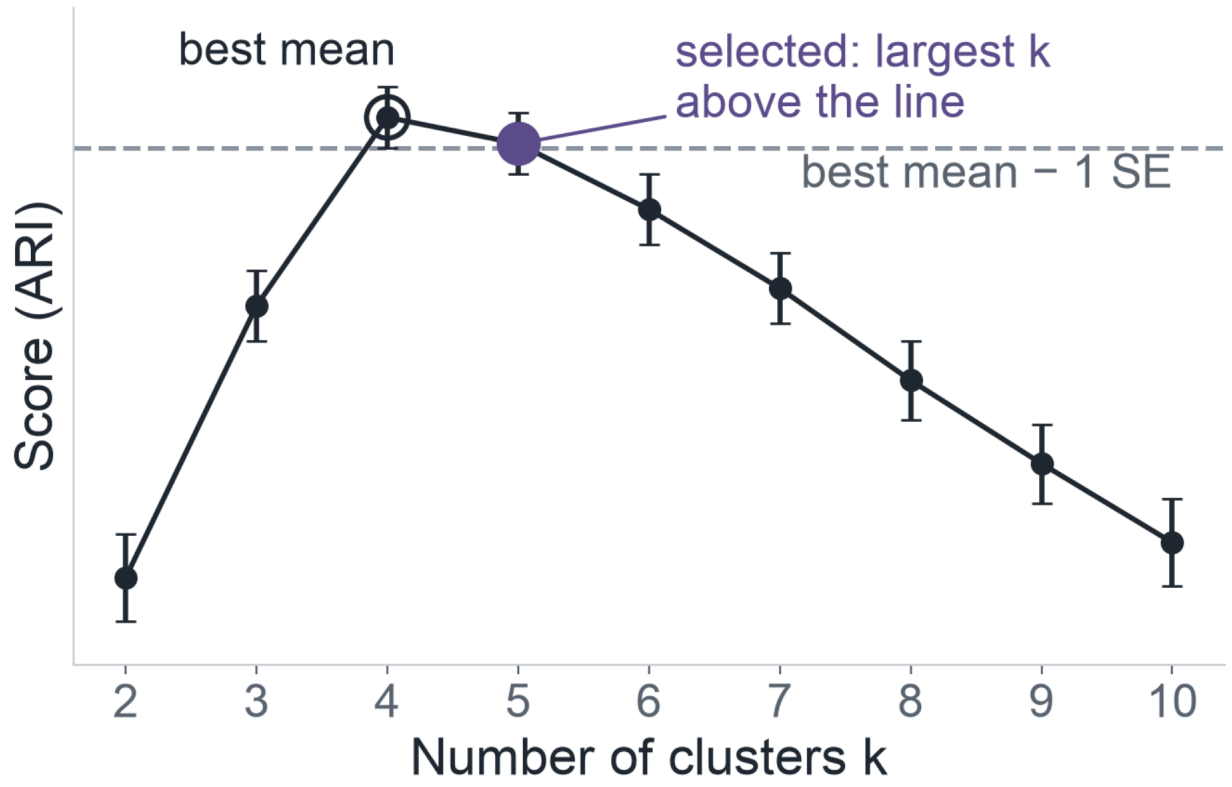


Illustration with made-up scores.

Find the candidate with the best mean score. Then take the most granular candidate (the largest k) whose mean is within one standard error of that best mean.

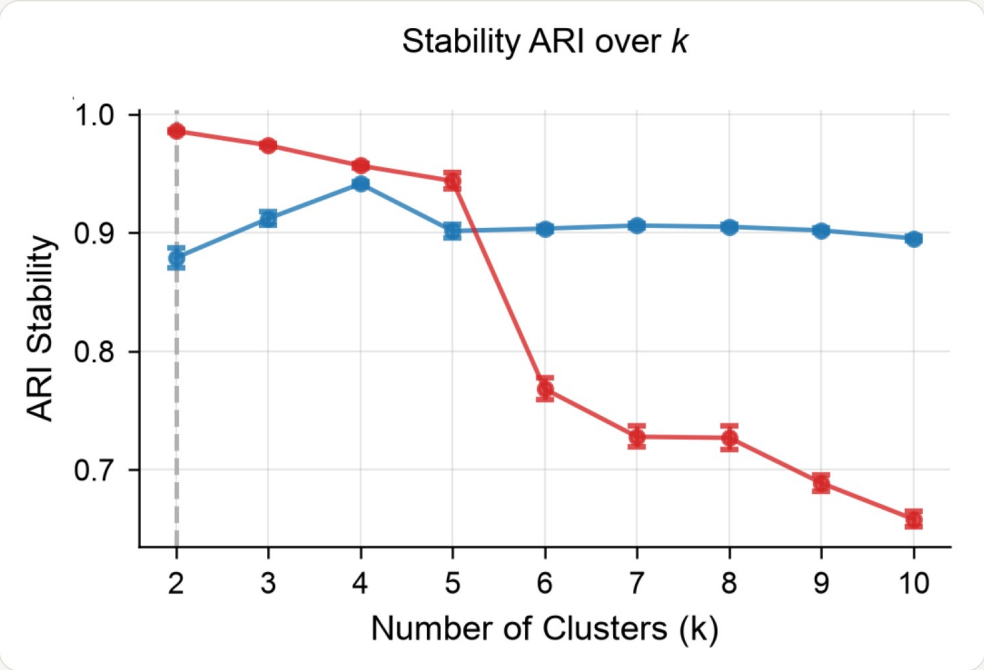
The idea comes from cross-validation in supervised learning.
When the scores are too close to tell apart, it picks the finer clustering.

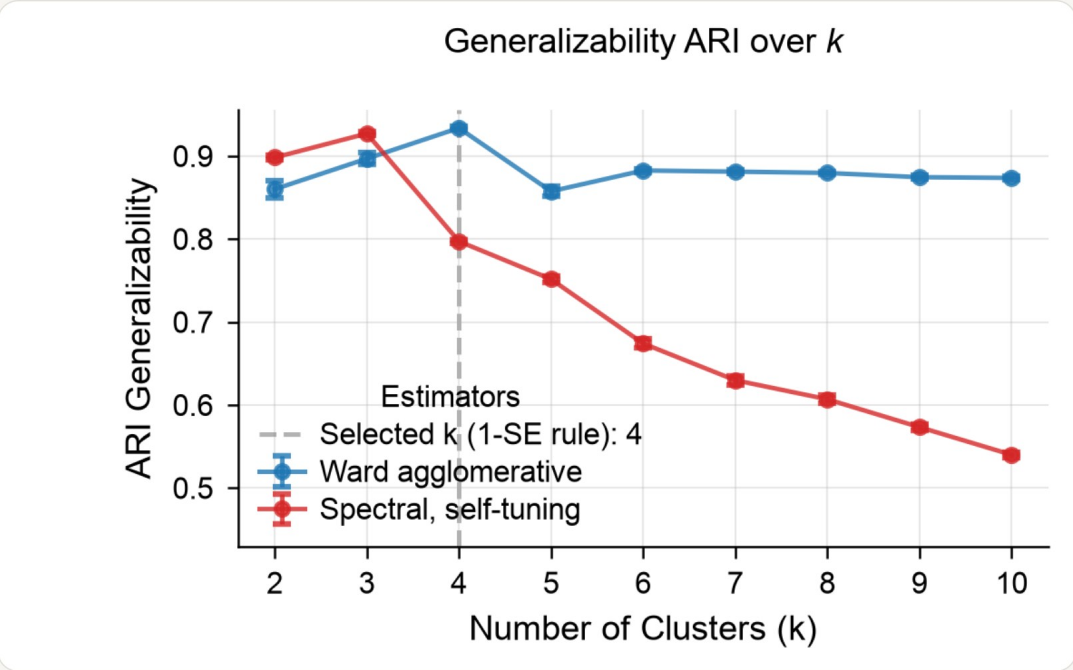
CARVE's default is stability with this rule. The maximum and quantile rules are also available.

Hastie, Tibshirani & Friedman, The Elements of Statistical Learning, 2009

READING THE OUTPUT · GLOBAL

Global: one curve per algorithm

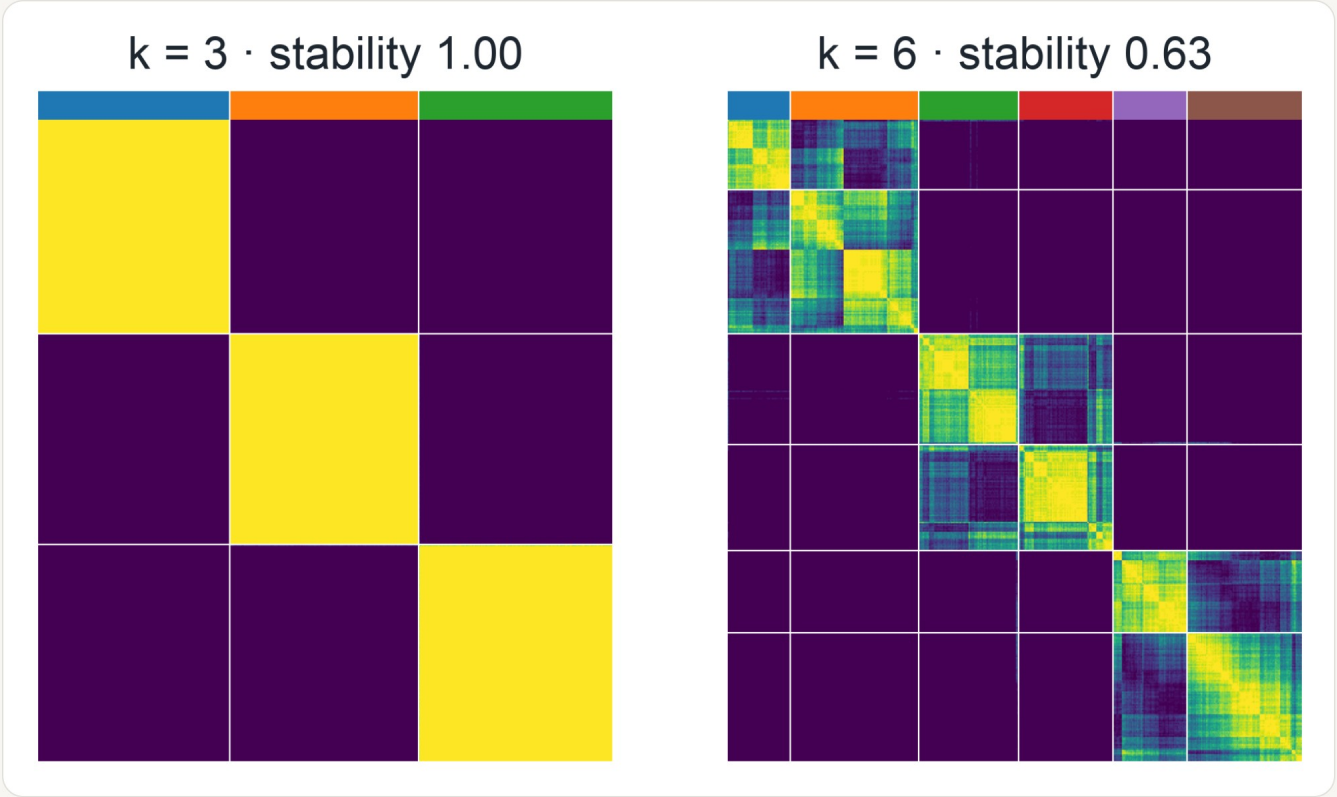




Klein data. Here the line colors are algorithms: Ward (blue) and spectral clustering (red), mean ± 1 standard error over 100 resamples. The dashed lines mark the 1-SE choice: spectral at k = 2 for stability (left), Ward at k = 4 for generalizability (right).

READING THE OUTPUT · CONSENSUS MATRIX

How often two cells end up together



One row and one column per cell. Each entry is the fraction of resamples that put the two cells in the same cluster, out of the resamples that drew both.

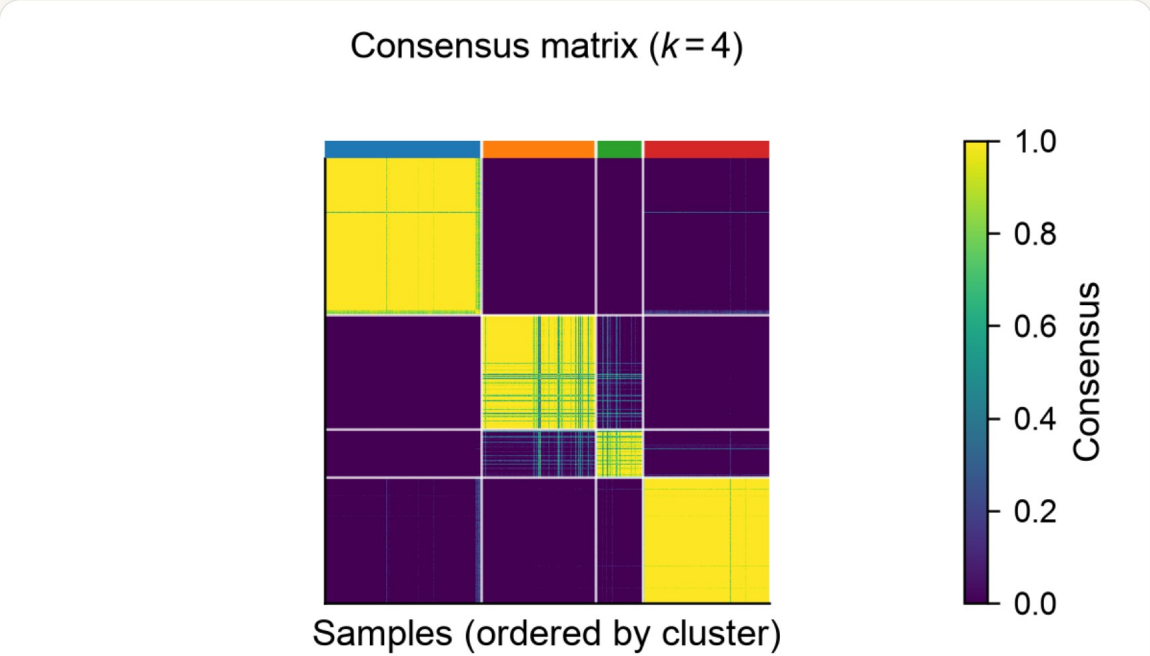
Yellow: always together. Dark: never together. Cells are sorted by cluster.

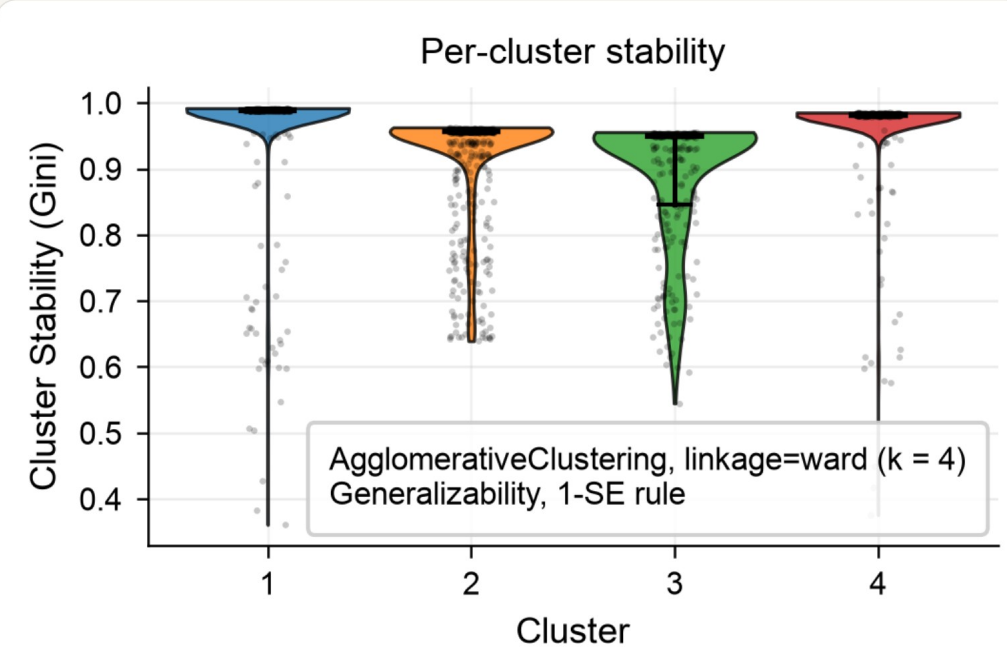
The three groups from the stability slide, k-means, 100 resamples.

k = 3 gives three solid blocks. At k = 6 each group is cut in two, the cut moves between resamples, and the blocks blur.

READING THE OUTPUT · CLUSTER LEVEL

Cluster level: which clusters hold up

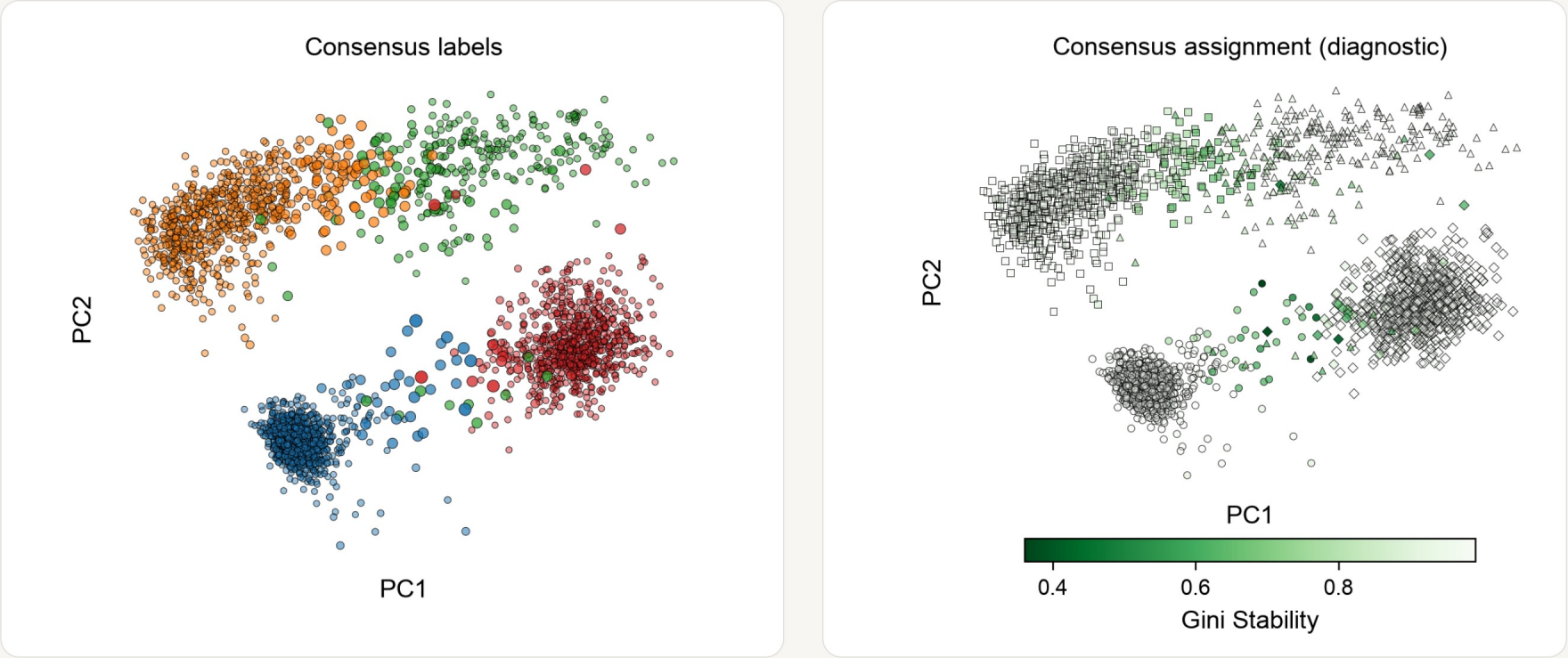




Ward, k = 4, the solution chosen by generalizability. Left: cells ordered by cluster; yellow means a pair of cells was almost always clustered together. Right: Gini stability of each cell, by cluster. Clusters 2 and 3 have long tails of unstable cells.

READING THE OUTPUT · CELL LEVEL

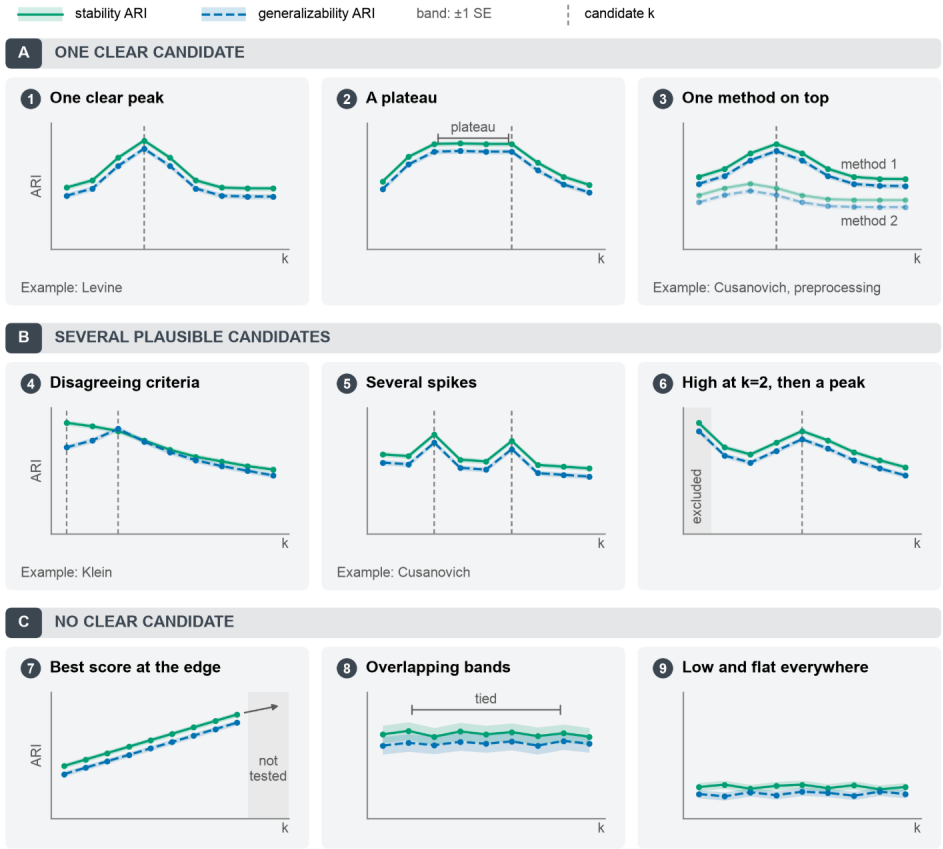
Cell level: which cells switch clusters



PCA of the Klein cells. Left: consensus clusters, less stable cells drawn larger. Right: marker shape is the cluster, color is Gini stability; dark green cells switch clusters between resamples. They sit on the borders between clusters.

READING THE OUTPUT

Common curve patterns and next steps



Row A, one clear candidate: a single peak, a plateau, or one method above the others. On a plateau, take the most granular solution.

Row B, several plausible candidates: the criteria disagree, the curves have several peaks, or k = 2 scores highest. Compare the candidates with the cluster and cell plots and with what you know about the biology.

Row C, no clear candidate: the best score sits at the edge of the range, the bands overlap, or everything is low. Widen the range, try other algorithms, and be cautious with any solution scoring below 0.5.

Schematic from the practical guidance in the CARVE manuscript's supplement

READING THE OUTPUT · CLUSTER LEVEL

Stability and generalizability as darts

Darts grouped tightly: the cluster is stable.

Darts on the bullseye: the cluster is generalizable.

The violins show the cells of cluster C2: stability on the left in green, generalizability on the right in blue.

Read each cluster's violins against these four cases.

Schematic from the practical guidance in the CARVE manuscript's supplement

CARVE · PREPROCESSING

Randomized preprocessing: one pipeline per resample

1. List the preprocessing pipelines you want to compare.
2. In each resample, draw one pipeline. The draws are split evenly over the list.
3. Fit the drawn pipeline separately on both subsamples and on the test set, then cluster and score as before.
4. After all resamples, group the scores by pipeline.

A pipeline whose output changes from one subsample to the next gets low stability and generalizability.

Pipelines in the atlas

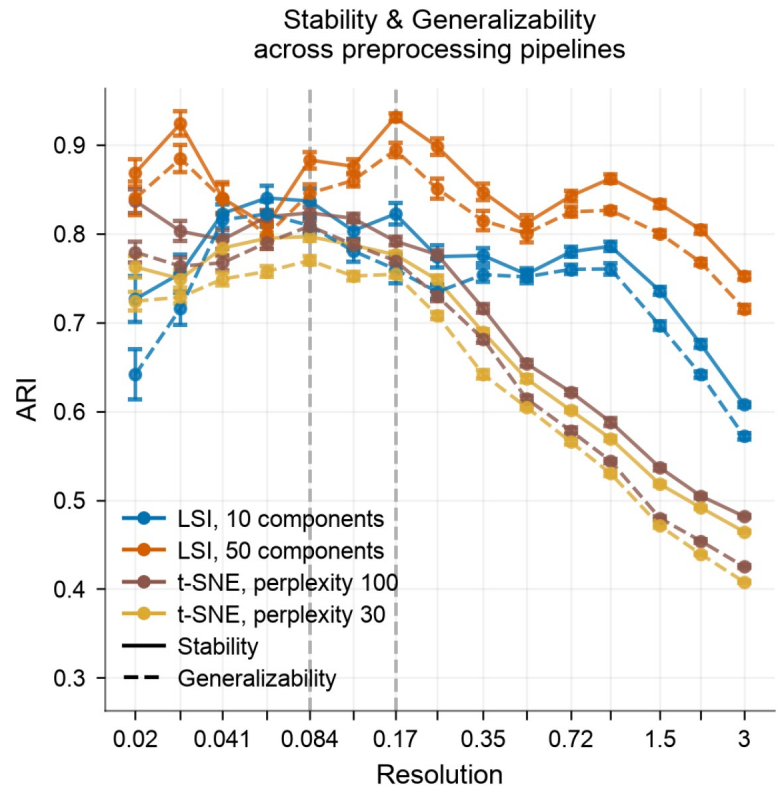
- LSI, 50 components
- LSI, first 10 components
- t-SNE of the LSI, perplexity 30
- t-SNE of the LSI, perplexity 100

The classifier is always trained on the 50-component LSI, because a t-SNE cannot place new cells.

Cusanovich et al., Cell 2018 · setup from the CARVE manuscript

READING THE OUTPUT · PREPROCESSING

Comparing preprocessing choices



Solid lines: stability. Dashed: generalizability.

Cusanovich sci-ATAC-seq atlas: 81,173 cells, Louvain clustering at 15 resolutions from 0.02 to 3.

In each resample CARVE draws one of four inputs: LSI with 50 or 10 components, or a t-SNE of the LSI with perplexity 30 or 100.

The 50-component LSI (orange) scores highest at the selected resolution, 0.17, and near the 30 clusters the authors reported. Both t-SNE inputs score lower.

Cluster on the LSI. Use t-SNE only to draw the cells.

Cusanovich et al., Cell 2018 · figure from the CARVE manuscript

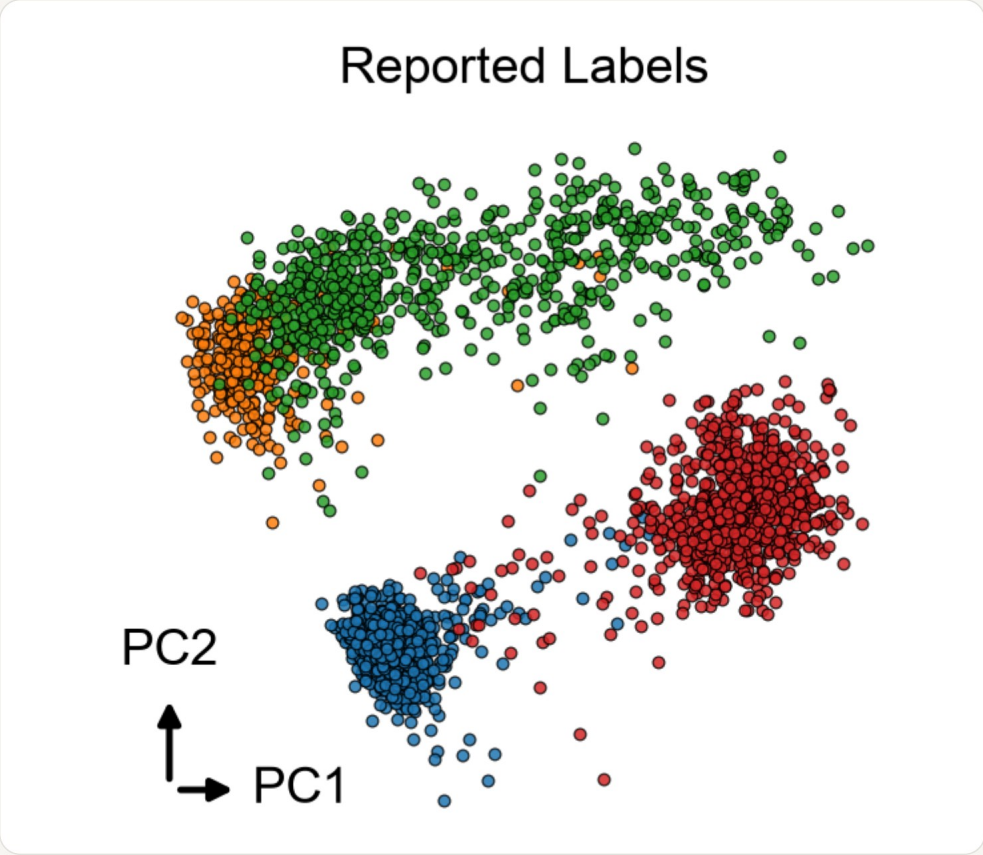
05

Case study: stem cells

Klein et al., Cell 2015 · droplet single-cell RNA-seq

CASE STUDY · DATA

Stem cells differentiating over seven days



Klein et al. (2015) withdrew leukemia inhibitory factor (LIF) from mouse embryonic stem cells and sequenced single cells on days 0, 2, 4 and 7.

We use all 2,717 cells and 2,000 highly variable genes. The collection day is used only to check the results.

- Day 0 · 933 cells
- Day 2 · 303 cells
- Day 4 · 683 cells
- Day 7 · 798 cells

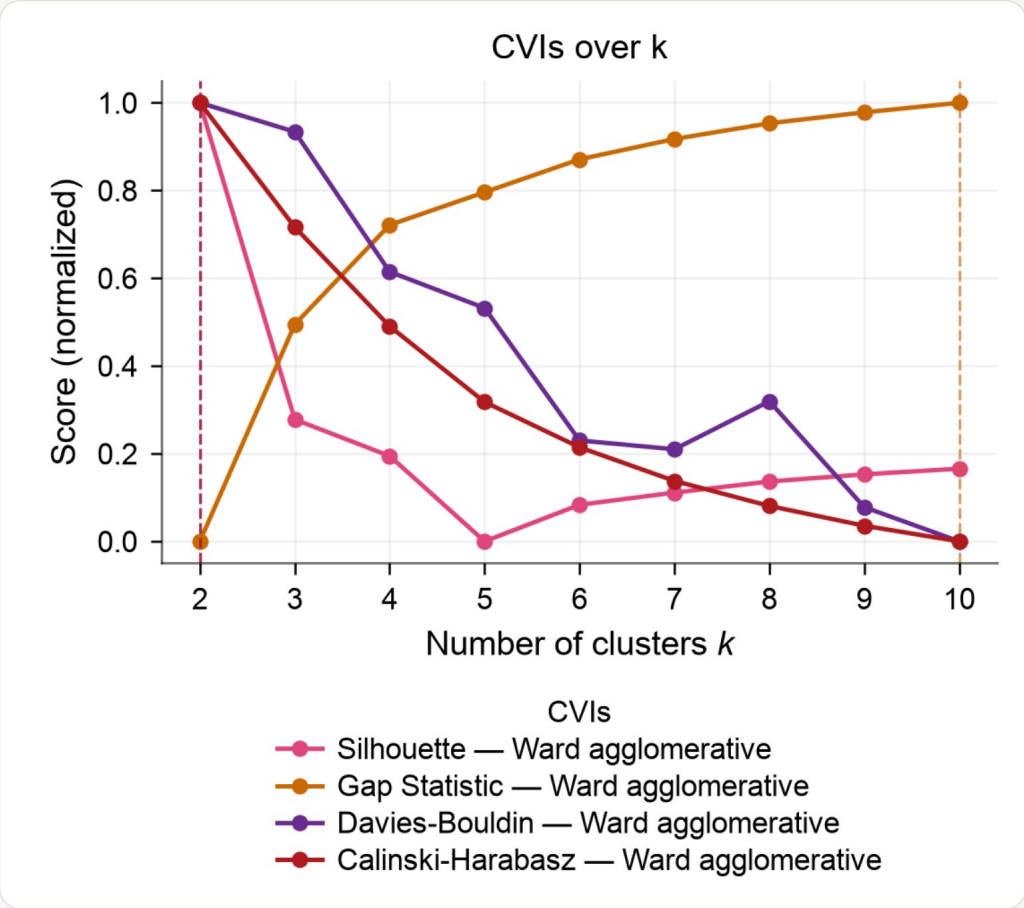
Days 2 and 4 overlap, because cells start to differentiate at different times.

CARVE run: Ward and spectral clustering, k = 2 to 10, 100 resamples, subsample fraction 0.618.

Klein et al., Cell 2015 · preprocessing as in the CARVE manuscript's supplement

CASE STUDY · VALIDITY INDICES

The indices pick k = 2 or k = 10



Ward clustering of 2,717 mouse embryonic stem cells, k = 2 to 10. Each index is rescaled to the range 0 to 1, higher is better.

Silhouette, Davies–Bouldin and Calinski–Harabasz pick k = 2.

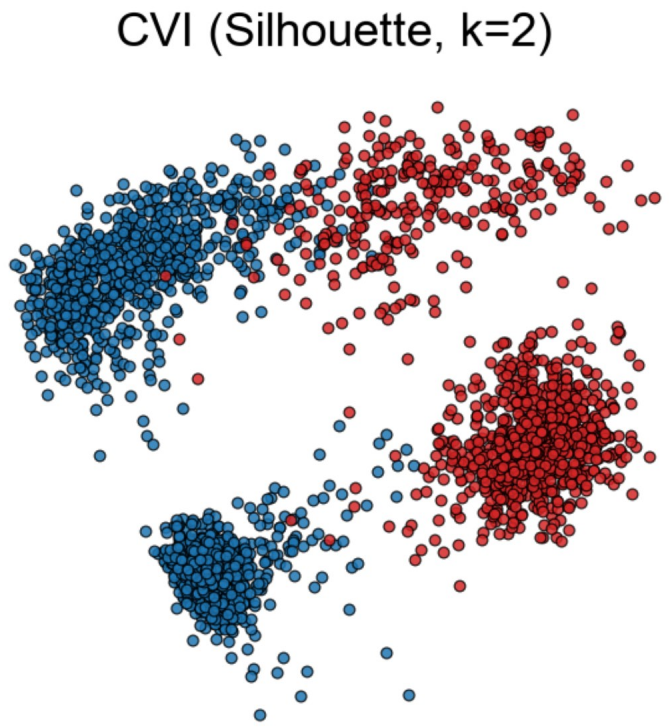
The gap statistic keeps rising and picks k = 10, the largest value we tried.

Three of the four indices pick two clusters. The next slide shows that split.

Klein et al., Cell 2015 · figure from the CARVE manuscript

CASE STUDY · VALIDITY INDICES

Silhouette picks two clusters



Each cluster drawn in the color of its closest collection day.

Silhouette, Davies-Bouldin and Calinski-Harabasz all choose Ward clustering with k = 2.

One cluster holds the day 0 and day 2 cells and part of day 4. The other holds day 7 and the rest of day 4.

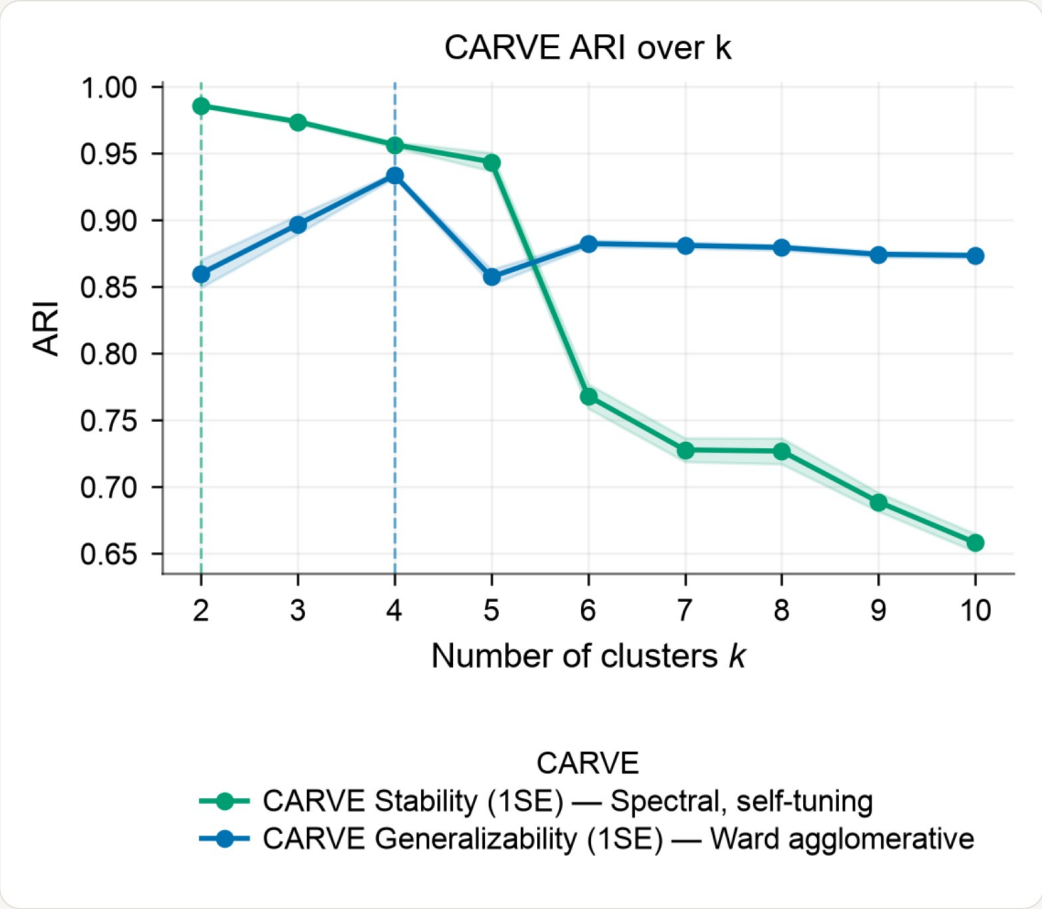
Agreement with the collection days: ARI 0.365.

M3C, a consensus clustering tool with its own rule for k, also picks k = 2 (ARI 0.368 on a 1,000-cell subsample).

M3C: John et al., Scientific Reports 2020

CASE STUDY · CARVE

Generalizability peaks at four clusters



Stability, shown for spectral clustering, its best algorithm, is highest at k = 2 and falls as k grows. There is no peak to choose.

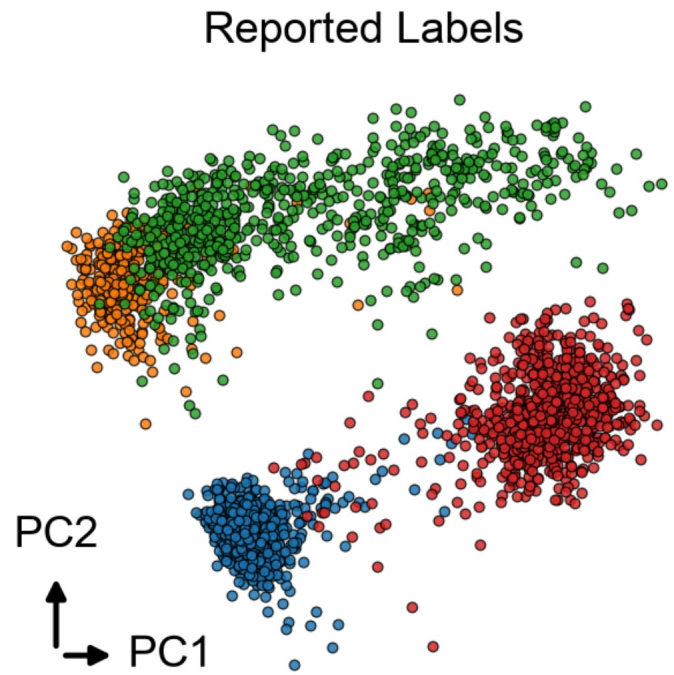
Generalizability, shown for Ward clustering, rises to a clear maximum at k = 4. The 1-SE rule selects it.

The criteria disagree, so we look at both solutions. k = 2 is the split from the previous slide. k = 4 is new.

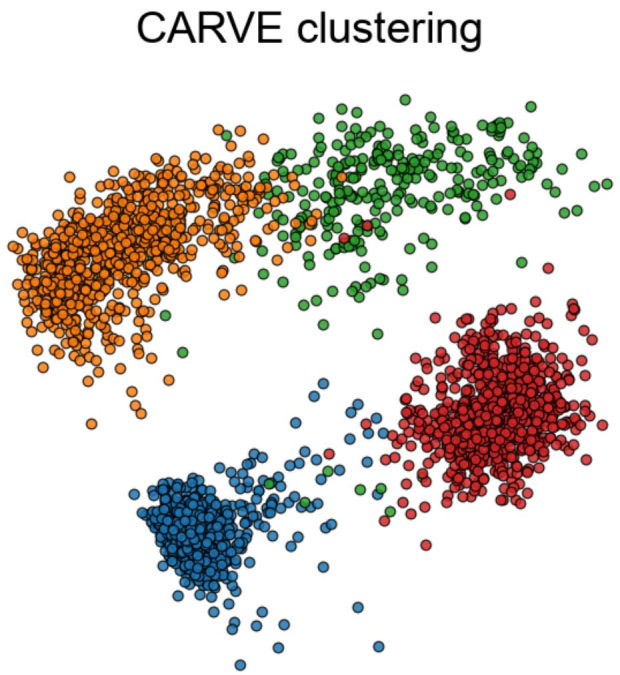
CASE STUDY · CARVE

Comparing CARVE's and Reported Labels

Collection day



CARVE: Ward, k = 4, consensus labels



0.791

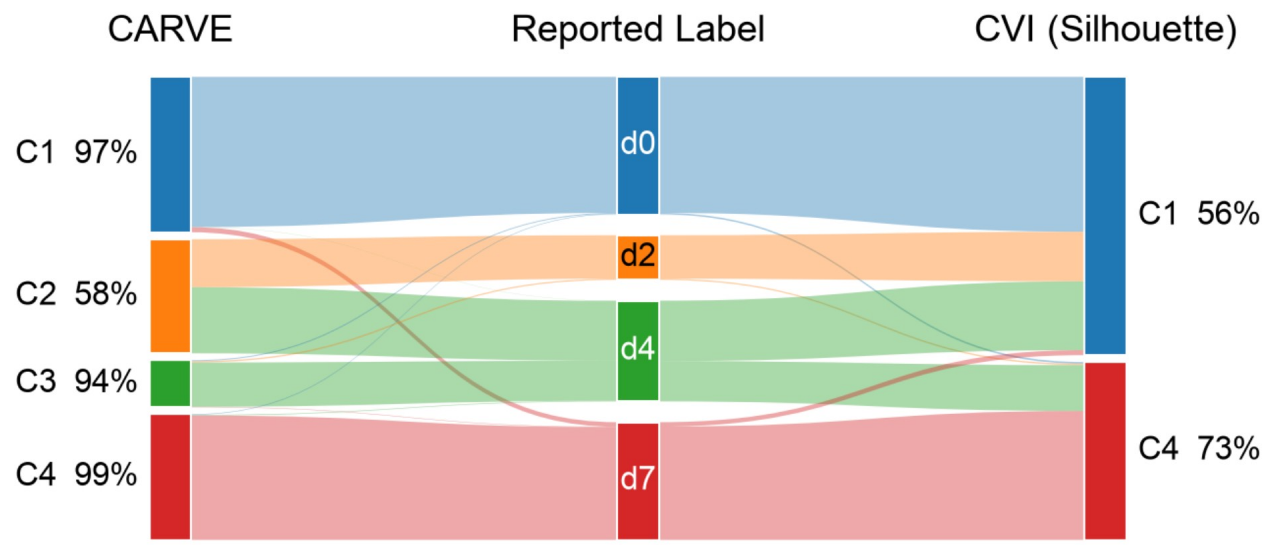
ARI with the reported cell labels

Silhouette at k = 2: 0.365

Each CARVE cluster is drawn in the color of the day it matches best.

CASE STUDY · CARVE

Days 2 and 4 are split across two clusters



Left: CARVE clusters with their purity, the share of cells from the most common day. Middle: collection day. Right: Silhouette clusters.

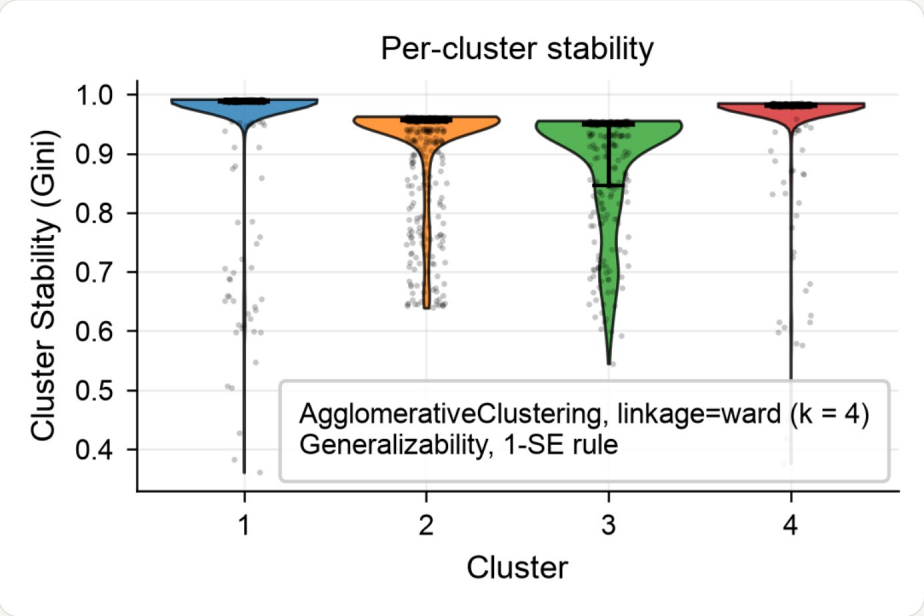
Cluster 2 holds 295 day 2 cells and 406 day 4 cells.

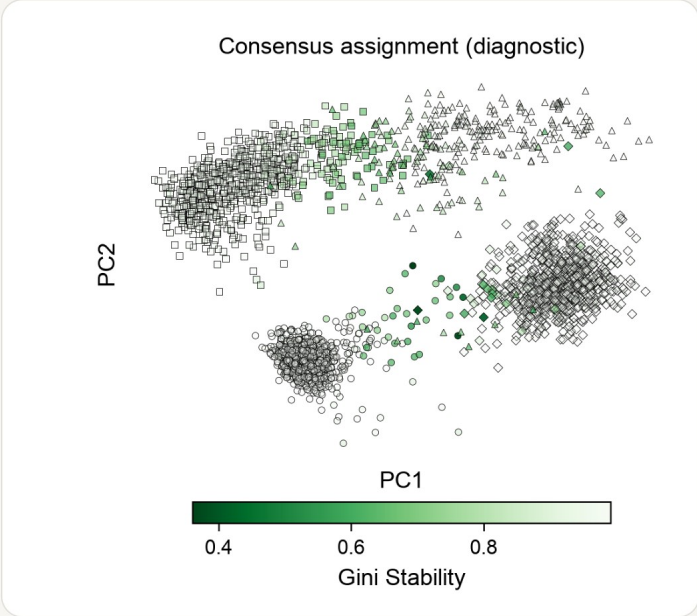
Cluster 3 holds 272 of the remaining 277 day 4 cells.

This matches what Klein et al. reported: after LIF withdrawal, cells start to differentiate at different times.

CASE STUDY · CARVE

Unstable cells sit between clusters





Mean Gini stability is 0.98 for the day 0 cluster and 0.97 for day 7, but 0.92 and 0.89 for the two clusters that mix days 2 and 4. The unstable cells sit on the borders: inside the upper arc and along the bridge between the two lower groups.

The four clusters are segments of one differentiation trajectory, not four separate cell types.

CASE STUDY · SUMMARY

Klein: chosen k and agreement with days

Method	Chosen k	ARI with days
CARVE generalizability, 1-SE rule (Ward)	4	0.791
Gap statistic (Ward)	10	0.741
Silhouette, Davies–Bouldin, Calinski–Harabasz (Ward)	2	0.365
M3C (1,000-cell subsample)	2	0.368

The gap statistic picked the largest k we tried, so its answer depends on the range we gave it. CARVE's k = 4 is a peak inside the range, and CARVE also shows which of its cells are uncertain.

USING CARVE

Running CARVE in Python

```
git clone https://github.com/DataSlingers/CARVE && pip install -e CARVE
```

```
import numpy as np
from sklearn.cluster import AgglomerativeClustering
from carve import CARVE, SpectralClustering

ks = np.arange(2, 11)
grids = [
    (AgglomerativeClustering, {"n_clusters": ks, "linkage": ["ward"]}),
    (SpectralClustering, {"n_clusters": ks, "affinity": ["self_tuning"]}),
]
carve = CARVE(estimator_param_grids=grids, n_resamples=100, random_state=0)
carve.fit(X)  # X: cells by features, e.g. 50 principal components

labels = carve.get_labels(measure="g", rule="1se", not_two=True)
carve.plot_metric_over_n_clusters(measure="g", rule="1se", not_two=True)
```

Full walkthrough: notebooks/Tutorial.ipynb in the repository

USING CARVE

Running CARVE in R

```
remotes::install_github("DataSlingers/CARVE", subdir = "carve-r")
```

```
library(CARVE)

grids <- list(
  estimator_grid(AgglomerativeClustering, n_clusters = 2:10, linkage = "ward"),
  estimator_grid(SpectralClustering, n_clusters = 2:10, affinity = "self_tuning")
)
fit <- carve(X, estimator_param_grids = grids, random_state = 0)

labels <- get_labels(fit, measure = "generalizability", rule = "1se", not_two = TRUE)
plot_metric_over_n_clusters(fit, measure = "generalizability", rule = "1se", not_two = TRUE)
```

Full walkthrough: notebooks/R_Tutorial.Rmd in the repository, including a Seurat workflow

IN PRACTICE

Recommendations for your own data

1. Try more than one algorithm, over a wide range of k or resolution.
2. Read the stability and generalizability curves together. If they disagree, look at both solutions.
3. Be suspicious of k = 2. Exclude it (not_two) unless two groups is a real hypothesis.
4. Check the cluster and cell plots before you name clusters. Unstable cells often mark a continuum.
5. Use what you know about the cells (marker genes, expected cell types, the design of the experiment) to choose among the candidates that score well.
6. Score your preprocessing choices too. A default is still a choice.
7. If nothing scores well, the data may not have clear groups. That is worth reporting.

SUMMARY

- Clustering results depend on the preprocessing, the algorithm, its settings and k, and there are no labels to check them against.
- Validity indices reward round, separated groups and often pick too few clusters on single-cell data.
- Stability and generalizability ask whether clusters come back on new subsamples and can be predicted for new cells.
- CARVE computes both for a grid of candidates, at the global, cluster and cell level.
- The scores narrow down the candidates. What you know about the cells decides among them.

Questions

Code, tutorials and the paper: github.com/DataSlingers/CARVE

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